Tragedy of the Confident Agent: Forced Exploration for Survival in Drifting Environments

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Abstract

Standard exploration-exploitation theory says: explore when uncertain. We show this advice is structurally incomplete in drifting environments — *sufficiently confident* agents cannot maintain bounded tracking unless they actively probe, a second exploration drive opposite-signed to the standard one and operating at *low* model uncertainty. A Lyapunov-survival argument establishes the mechanism: as a confident agent's update gain falls below the rate at which reality drifts beneath it, the maximum tolerable observation noise collapses to zero. The scalar formulation admits a "blank-wall" degenerate update — an action maximizing scalar information yield in directions orthogonal to drift. We resolve this via a Linear Matrix Inequality on the Fisher Information Matrix with a matrix-valued Lagrange multiplier $\Lambda \succeq 0$ supported on the drifting subspace, whose range-containment hypothesis $\mathrm{range}(\Lambda) \subseteq \mathrm{range}(\mathcal{I}_{\min})$ denies blank-wall updates any LMI-survival reward; directional discrimination rests on anisotropic drift. A diagonal 2D Gaussian simulation demonstrates directional discrimination: the matrix-LMI controller selects drift-aligned probing actions and survives at 100% across drift levels (at zero pragmatic reward, a survival-vs-reward bistability of the discrete action set) where scalar surrogates collapse to 0% via blank-wall degenerate updates; the experiment validates directional discrimination and finite-horizon survival, not a smooth Pareto frontier. In drifting tracking environments, the framework gives a *structural* (Lyapunov-persistence) obstruction to the dark-room failure mode rather than a parametric (prior-tuning) one — a dark-room agent has $\mathbb{E}[\mathcal{I}_o(a)] = 0$ in the drifting subspace and violates the survival LMI. The "tragedy" is the *structural irony* of the inversion itself: an agent is mathematically required to abandon exploitation and probe its environment precisely when its internal model tells it that it knows everything.

1 Introduction

The default exploration-exploitation contract in reinforcement learning, active inference, and Bayesian decision-making is this: an agent should gather information when it is *uncertain* about its environment and exploit when it is *confident*. Epistemic bonuses scale positively with model uncertainty — UCB-style confidence bounds [1], intrinsic-curiosity prediction-error rewards [2], variational information maximization [3], random network distillation [4], and the epistemic value term in active inference's expected free energy [5] all share this monotonicity. As model uncertainty $U_M \downarrow 0$, all such drives go to zero and the agent is supposed to settle into pure exploitation.

This paper shows that, in any environment whose state drifts with rate $\rho > 0$, the picture is incomplete. There is a *second* class of exploration drive — opposite-signed in U_M-dependence to the standard epistemic one — that the standard picture misses entirely. As $U_M \to 0$ the drive's *budget* — the maximum observation noise compatible with bounded tracking — collapses to zero, so any controller respecting the budget must intensify probing precisely when the agent is most confident. The drive is not a preference, prior, or curiosity term: it is a *persistence imperative* that follows from coupling the agent's Bayesian update gain to a Lyapunov inequality on its tracking mismatch. Failing to obey it produces unbounded mismatch, regardless of how good the agent's model currently is.

More concretely: the Lyapunov persistence inequality $\alpha > \rho/R$ — coupling correction rate α to drift rate ρ, normalized by mismatch capacity R — combined with the agent's optimal Kalman gain produces a *survival ceiling* $U_o^{\max}(M_t) = U_M(Rc/\rho - 1)$ on observation noise. The bound has three regimes: feasible ($Rc > \rho$), where the ceiling is positive and collapses to zero as confidence rises; threshold ($Rc = \rho$), where only the singular $U_o = 0$ limit is admissible; and infeasible ($Rc < \rho$), where the agent's peak correction rate is below drift and exploration alone cannot rescue survival — a *sharper* structural obstruction than the feasible-regime collapse. We develop the feasible regime; in it, $U_M \to 0$ forces any agent satisfying the linear-Gaussian / sector-bounded conditions of Section 3.2 to seek pristine observations or lose bounded tracking.

In multi-dimensional environments the scalar form admits a "blank-wall" failure: actions whose observation channel is orthogonal to drift trivially satisfy the magnitude bound while contributing zero information about the drifting subspace. We resolve this by lifting to a Linear Matrix Inequality $\mathbb{E}_\pi[\mathcal{I}_o(a)] \succeq \mathcal{I}_{\text{min}}$ on the agent's expected Fisher Information Matrix, with a matrix Lagrange multiplier $\Lambda \succeq 0$ whose range-containment $\text{range}(\Lambda) \subseteq \text{range}(\mathcal{I}_{\text{min}})$ denies blank-wall actions any LMI-survival reward — directional discrimination by structure rather than magnitude, contingent on a controller-design hypothesis on Λ (satisfied by construction by $\Lambda \propto \mathcal{I}_{\text{min}}$).

Empirically, in a 2D Gaussian setup, a matrix-LMI controller selects the drift-aligned probe 100% across $\sigma_w \geq 1.5$ — and accumulates *zero* pragmatic reward in this regime, a survival-vs-reward bistability of the discrete action set discussed in Appendix B.5 — while scalar-surrogate controllers collapse to 0% via the blank-wall trap. At $\sigma_w \leq 1.0$ the survival drive is not yet activated and the LMI controller's selection is bit-identical to greedy (which also degrades with σ_w); the trap is decisive only in the $\sigma_w \geq 1.5$ regime where the persistence margin closes. The experiment validates *directional discrimination and finite-horizon survival*, not a smooth survival/reward Pareto frontier.

Why this gap matters Active inference's *dark-room problem*[6] — surprise minimization made trivially perfect by sensory deprivation — has a canonical parametric reply[7]: equip the agent with priors preferring engagement. Koudahl et al. 2021 [8] show this escape is fragile in the linear-Gaussian state-space setting — the most analytically tractable corner of active inference — where the epistemic component of expected free energy *collapses* to a state-independent constant, leaving only the pragmatic (prior-driven) term. Adaptive control has pointed at the underlying issue for four decades in different vocabulary: Anderson's *bursting*[9] (low persistent excitation → drift-along-unidentified-subspaces → sudden destabilization), Anderson's later explicit diagnosis[10] (continuous adaptation requires continuous identifying-information injection), dual-control[11–13] making the injection an objective, and directional Fisher-information experiment design[14–17] sharpening to *along-the-directions-that-matter*. These lineages contain all the constituent pieces of the structural escape but have never been composed into a single derivation yielding a *low-uncertainty* exploration drive forced by structural-stability considerations. To our knowledge, no published framework composes these lineages this way.

Contributions (1) Section 4 — a sector-Lyapunov persistence inequality (Lemma 3.1) couples to the Bayesian update gain to give the survival-noise threshold $U_o^{\max}(M_t) = U_M(Rc/\rho - 1)$, which collapses to zero as $U_M \to 0$, and a *survival-margin controller family* (Definition 4.1) — policies whose effective exploration weight diverges by design as the persistence margin closes (illustrative members: $\lambda \propto 1/U_M$, $\lambda \propto U_M/(R - R^*)$). (2) Section 4 — a blank-wall attack on the scalar form (actions orthogonal to drift trivially satisfy the magnitude bound while drifting subspace diverges); resolved by lifting to an LMI (Theorem 4.2) with matrix Lagrange multiplier $\Lambda \succeq 0$ whose range-containment hypothesis $\text{range}(\Lambda) \subseteq \text{range}(\mathcal{I}_{\text{min}})$ denies blank-wall actions any LMI-survival reward (Proposition 4.3). (3) Section 4.4 — a 2D Gaussian simulation in which a scalar margin-triggered controller is captured by a wall-extreme trap action while a matrix-LMI controller exclusively selects the drift-aligned probe and survives at 100% across the drift sweep, with LMI $\equiv$ greedy at low drift (the survival drive is silent when not needed). (4) Section 6.2 — a structural (Lyapunov-persistence) obstruction to the dark-room failure mode *in drifting tracking environments* rather than a parametric (prior-tuning) one, robust to the linear-Gaussian EFE-collapse. The argument's scope is honest about its assumptions: linear-Gaussian dynamics, the CIY $\propto 1/U_o$ surrogate (exact only in 1D Gaussian-linear), expectation-LMI not pathwise-controlling P_t without per-action structure, simultaneously-diagonalizable empirical setup with steady-state DARE-derived $\mathcal{I}_{\min}$ — each named in Section 6.

2 Related Work

We organize the related work by the four lineages identified in Section 1, with explicit cite-and-distinguish.

Adaptive control under drift The structural ancestor. Anderson 1985 [9] established the bursting phenomenon: under low persistent excitation, parameter drift along unidentified subspaces produces sudden destabilization even in nominally stable adaptive systems. Anderson 2005 [10] retrospectively diagnosed continuous adaptation as fundamentally requiring continuous identifying information. Kreisselmeier 1986 [18], Solo 1996 [19], and Bittanti and Campi 2000 [20] extended persistent-excitation conditions to time-varying parameter regimes. We adopt the structural diagnosis (continuous information injection is required) and the Lyapunov machinery (sector conditions, ultimate boundedness). We extend by deriving the *low-uncertainty* mandate from the same Lyapunov inequality, by giving a directional shadow price via the LMI lift, and by explicitly composing with the active-inference dark-room debate.

Dual control and probing The intentional-information ancestor. Bar-Shalom 1981 [11], Tse et al. 1973 [21], Filatov and Unbehauen 2000 [22], Mesbah 2017 [12], and Heirung et al. 2017 [13] develop probing-for-identification objectives. We adopt the structural insight that controllers must inject identifying information at a cost. We extend by deriving the probing requirement from a Lyapunov *survival* condition rather than a Bayesian-decision-theoretic trade-off, by tying the probing direction explicitly to the drift's eigenstructure, and by connecting to the active-inference literature.

Directional Fisher-information / LMI / experiment design The blank-wall-resolution ancestor. Hjalmarsson 2005 [14] established closed-loop experiment design with directional FIM constraints. Jansson and Hjalmarsson 2005 [23] gave the LMI form admitting frequency-wise specifications. Tanaka et al. 2017 [15] formulated Gaussian sequential rate-distortion as an SDP — the closest existing LMI/SDP work to our (6). Lee et al. 2019 [16] and Goel et al. 2020 [17] developed directional and variable-direction forgetting. We adopt the LMI machinery and the directional Fisher-information formulation. We extend by tying the LMI to *survival* (Lyapunov persistence) rather than estimation accuracy, and by deriving complementary slackness on direction as the mathematical resolution of the blank-wall attack.

Active-inference dark-room debate The active-inference framing we sidestep. Sun and Firestone 2020 [6] articulated the dark-room critique. Van de Cruys et al. 2020 [7] provide the canonical parametric reply. Koudahl et al. 2021 [8] empirically demonstrate that the EFE epistemic term collapses in linear-Gaussian state-space settings — making the parametric escape do all the work. Friston et al. 2015 [5] and the active-inference baseline [24–26] establish epistemic-value-as-exploration. We diverge structurally: the survival drive is forced by Lyapunov dynamics, not by preferences-as-priors, and is robust to the linear-Gaussian EFE-collapse.

Non-stationary bandits / RL Slivkins and Upfal 2008 [27], Garivier and Moulines 2008 [28], Russac et al. 2019 [29], and Min and Russo 2023 [30] treat non-stationarity via explore-more-when-uncertain bonuses; our structurally distinct term firing at *low* uncertainty under drift is complementary.

Concept drift, continual learning, open-world recognition Our bounded-rate-drift setting is a quantitatively-characterized special case of broader non-stationary-environment frameworks: *concept drift detection* [31], *continual / lifelong learning* [32], *open-world recognition* [33], and the closest direct neighbor — *Lyapunov-stable adaptive control for multimodal concept drift* [34], which shares the Lyapunov uniform-ultimate-boundedness framing but adapts *algorithmic* hyperparameters (learning rate, modality fusion weights) under classification drift rather than selecting state-perturbing control inputs against an action-conditional Fisher information matrix. Those literatures focus on detecting or adapting to novel categories or distribution shifts; we identify a *structural* exploration drive forced by the drift rate itself, with the LMI matrix lift handling the directional-discrimination subproblem under anisotropic drift.

Standard RL exploration / intrinsic motivation Auer et al. 2002 [1] (UCB), Pathak et al. 2017 [2] (ICM), Houthooft et al. 2016 [3] (VIME), Burda et al. 2019 [4] (RND), Schmidhuber 2010 [35] (compression-progress curiosity), Oudeyer and Kaplan 2007 [36] (typology of computational intrinsic motivation), and the broader intrinsic-motivation literature share the positive-with-uncertainty multiplier — the λ_{info} half of the dual-drive picture (20), complete in stationary environments and incomplete in drifting ones.

Safe RL and constrained policy optimization Constrained-MDP formulations [37] and constrained-policy-optimization [38] treat safety as a value-function-level constraint with a Lagrangian penalty; shielded RL [39] treats safety as a temporal-logic-filtered action layer. Our survival drive is the dual to a *Lyapunov persistence* constraint rather than a value-function-level safety constraint, and the blank-wall failure of scalar magnitude — actions satisfying the constraint without serving its purpose — is the same structural pattern these literatures address by direction-blind constraint, not by directional discrimination.

Information bottleneck and rate-distortion Tishby et al. 1999 [40], Tishby and Polani 2011 [41], and Cover and Thomas 2006 [42] — (20)'s dual-drive structure fits the IB framing with a drift-coupled directional acquisition constraint added to the standard compression-prediction trade-off.

3 Setup

We work in continuous-time mismatch dynamics with a discrete-action policy. Worst-case-sharp results are derived in *Model D* (deterministic-bounded disturbance $\|w\| \leq \rho$); the corresponding *Model S* (stochastic Gaussian disturbance, $\mathbb{E}\|w\|^2 = \sigma_w^2$) results are mean-square or finite-horizon high-probability and are used in the simulation of Appendix H. Both disturbance models are formalized in Section 3.2; ceiling and threshold formulas appear in Model D unless noted, with Model S analogs given inline at first appearance.

3.1 Agent state

Let Ω_t be the unobserved environmental state and M_t the agent's belief (point estimate plus uncertainty). The mismatch signal is $\delta_t = M_t - \Omega_t$. We track:

- **Model uncertainty** $U_M(t) \geq 0$: the agent's posterior variance over Ω_t.
- **Observation noise** $U_o(a) \geq 0$: action-dependent variance of the agent's observation $o_t = \Omega_t + \nu_t$, $\nu_t \sim \mathcal{N}(0, U_o(a))$.

The agent's optimal Bayesian update gain is the Kalman value

$$\eta^* = \frac{U_M}{U_M + U_o}. \tag{1}$$

3.2 Mismatch dynamics under drift

We adopt the standard formulation in which mismatch evolves as a sector-bounded correction process driven by environmental drift:

$$\frac{d\delta}{dt} = -F(\delta) + w(t), \tag{2}$$

where $w(t)$ is the disturbance process. We consider two disturbance models, following Khalil 2002 [43] ch. 4 & 9:

- **Model D** (deterministic-bounded): $\|w(t)\| \leq \rho$ for all t.

- **Model S** (stochastic isotropic Gaussian): each coordinate of $w(t)$ has variance σ_w^2, so $\mathbb{E}\|w\|^2 = n\sigma_w^2$ and $\mathrm{Cov}(w_t) = \sigma_w^2 I_n$. The matrix drift covariance used in §4 is $Q_\rho = \mathrm{Cov}(w_t)$ (which becomes $\mathrm{diag}(\sigma_w^2, 0, \ldots, 0)$ when only one coordinate drifts, as in the Appendix H simulation).

The correction F satisfies the local sector condition for $\|\delta\| \le R$:

$$\delta^\top F(\delta) \ge \alpha\,\|\delta\|^2, \quad \alpha > 0. \tag{3}$$

The constant α is the *effective correction rate*. For Bayesian-update agents, $\alpha = \eta^* \cdot c$ where c is the worst-case directional fidelity of the observation channel (derivation in Appendix F.1). The constant $R > 0$ is the *model-class capacity*: the largest mismatch the correction function can still respond to before saturating.

3.3 Persistence

A standard Lyapunov argument[43] ch. 9 — sector-condition machinery from[44] — yields:

Lemma 3.1 (Persistence threshold; Model D, robust form)**.** *Under* (2)–(3) *with bounded-disturbance class $\mathcal{W} = \{w : \|w(t)\| \le \rho\ \forall t\}$:*

(i) Sufficient (forward, all sector-bounded F). *If $\alpha > \rho/R$, then for every $w \in \mathcal{W}$ and every $\delta(0) \in \mathcal{B}_R$ the trajectory stays in $\mathcal{B}_R$ and is ultimately bounded by $R^* = \rho/\alpha$.*

(ii) Worst-case sharp (reverse, with finite exit-time witness). *Let F additionally satisfy the upper sector bound $\delta^\top F(\delta) \le (\alpha+\gamma)\|\delta\|^2$ on $\mathcal{B}_R$ for some sector slack $\gamma \ge 0$. If $\alpha + \gamma < \rho/R$, the adversarial disturbance $w^*(t) = \rho\,\delta(t)/\|\delta(t)\|$ — admissible since $\|w^*\| = \rho$ — drives every trajectory in $\mathcal{B}_R \setminus \{0\}$ out of $\mathcal{B}_R$ in finite time bounded by*

$$T_{\text{exit}}(\delta_0) \;\le\; \frac{1}{\alpha+\gamma}\,\log\frac{\rho/(\alpha+\gamma) - \|\delta_0\|}{\rho/(\alpha+\gamma) - R}. \tag{4}$$

(iii) Tight-sector / linear-correction (universal-trajectory iff). *If $F(\delta) = M\delta$ (equivalently $\gamma = 0$ in (ii); this is the regime Appendix F onward uses via the Kalman-gain bridge $\alpha = \eta^* c$), then under worst-case $w \in \mathcal{W}$,*

$$\boxed{\alpha > \frac{\rho}{R}} \tag{5}$$

is two-sided and universal-trajectory, with exit-time bound (4) *at $\gamma = 0$.*

Proof. Sector-Lyapunov ultimate boundedness gives positive invariance of $\mathcal{B}_R$ above threshold (i); a worst-case-disturbance comparison-principle argument against the linear ODE $\dot\mu = -(\alpha+\gamma)\mu + \rho$ produces the finite-time exit bound below threshold (ii); the linear-correction substitution $\gamma = 0$ collapses the iff (iii). Full argument in Appendix A.2. □

The Model S analog is *mean-square*: $R_S^* = \sigma_w\sqrt{n/(2\alpha)}$ with stationary-second-moment boundedness $\mathbb{E}\|\delta\|^2 \le R^2$ requiring $\alpha > n\sigma_w^2/(2R^2)$. This is *not* pathwise infinite-horizon persistence — continuous Gaussian noise eventually exits any bounded set with probability 1 — so Model S "persistence" in Appendix F onward should be read as mean-square boundedness or finite-horizon high-probability survival; pathwise infinite-horizon $T_{\text{exit}} = \infty$ is a Model D claim.

Above threshold, no admissible disturbance drives the agent out of $\mathcal{B}_R$; below threshold, the worst-case disturbance does so in the finite time (4). The threshold is a *cliff in the worst-case guarantee structure*: above it, $T_{\text{exit}} = \infty$; below it, T_{exit} is finite and shrinks rapidly as the agent drifts deeper sub-threshold.

We adopt the sector-Lyapunov machinery, optimal-Kalman gain (1), DARE-form steady-state covariance, and information-form FIM additivity from standard control theory[43–45]. The novel content — *coupling* (1) and (5) via the Bayesian update gain into a directional survival mandate — is stated in Section 4 and proved in Appendix F–Appendix G.

4 Main Results

We characterize policies that maintain bounded tracking under drift, give a sufficient condition on the agent's expected Fisher Information Matrix for survival, and show that the matrix Lagrangian discriminates by direction rather than by aggregate magnitude. Three load-bearing claims: a controller-family construct (Definition 4.1), a sufficient LMI (Theorem 4.2), and a directional kicker (Proposition 4.3).

4.1 The survival-margin controller family

The Lyapunov persistence inequality $\alpha > \rho/R$ requires the agent's effective correction rate α to outpace the drift rate ρ normalized by mismatch capacity R. For a Bayesian-update agent the bridge $\alpha = \eta^* c$ holds — the optimal Kalman gain η^* from (1) times the worst-case directional channel sensitivity c — so the persistence inequality fails as $U_M \to 0$ unless the agent actively reduces U_o along the drifting subspace. We capture the policies that respond to this Lyapunov mandate by construction:

Definition 4.1 (Survival-margin controller family)**.** A policy weight $\lambda_{\text{surv}} : M_t \mapsto \mathbb{R}_{\geq 0}$ belongs to the *survival-margin controller family* if there exists a *persistence margin* $\Delta : M_t \mapsto \mathbb{R}_{\geq 0}$ — any continuous functional vanishing exactly when (5) becomes critical (e.g., $\Delta = U_M$ for the confidence margin, $\Delta = R - R^*$ for the dynamic mismatch margin where $R^* = \rho/\alpha$ is the agent's current ultimate-bound radius) — such that (i) $\lambda_{\text{surv}}(M_t) \to \infty$ as $\Delta(M_t) \to 0^+$, and (ii) $\lambda_{\text{surv}}(M_t)$ is finite when $\Delta(M_t) > 0$ is bounded below.

Two natural family members are illustrative: the *inverse-uncertainty* form $\lambda^{(1)}_{\text{surv}}(M_t) = k/U_M$ (reading $\Delta = U_M$), which diverges as the agent's confidence rises; and the *margin-triggered* form $\lambda^{(2)}_{\text{surv}}(M_t) = k\, U_M/(R - R^*)$ (reading $\Delta = R - R^*$), which diverges as the dynamic mismatch margin closes. Family membership is a *qualitative* divergence property; whether a chosen member then enforces the survival constraint requires four explicit conditions formalized as Proposition A.4 (F-A-G-P: strict feasibility, drift-aligned bonus, gain dominance, argmax expressibility), with full discussion in Appendix A.7.

4.2 The averaged-information LMI

The natural multi-dimensional rigorous form is not an algebraic lift of $\mathbb{E}_\pi[U_o(a)] \leq U_o^{\max}$ — the scalar observation-noise expectation form is a Jensen-strong sufficient surrogate (Appendix G.6) that coincides with the information-yield form only on degenerate or constant-noise supports. The rigorous form is a matrix lower bound on the action-conditional Fisher Information Matrix $\mathcal{I}_o(a) = H^\top R_o(a)^{-1} H$ — exactly the matrix-valued causal information yield in the Gaussian-linear setting, entering the information-form Kalman recursion additively. It gives a *steady-state-covariance* survival floor; per-step pathwise survival requires per-action strengthening (the Pathwise vs. expectation gap remark below):

Theorem 4.2 (Averaged-information LMI sufficient condition)**.** *Assume the linear-Gaussian model* (21) *with detectable* (A, H)*, stabilizable* $(A, Q_\rho^{1/2})$*, and a stationary randomized policy* π *inducing i.i.d. action draws. The deterministic DARE driven by the expected observation Fisher matrix* $\mathbb{E}_\pi[\mathcal{I}_o(a)]$ *admits a stabilizing steady-state solution* Σ_δ *with* $\lambda_{\max}(\Sigma_\delta) < R^2$ *if*

$$\boxed{\mathbb{E}_{a\sim\pi}\big[\mathcal{I}_o(a)\big] \;\succeq\; \mathcal{I}_{\min}(Q_\rho, A, R^2)} \tag{6}$$

where $\mathcal{I}_{\min} \succeq 0$ *is the Fisher-information floor for the averaged DARE*[45] *§4.4. In the simultaneously-diagonalizable case* $\mathcal{I}_{\min}$ *has a closed-form per-eigendirection expression (Appendix A.3); in the general non-commuting case* (6) *is an SDP feasibility floor rather than a unique tight floor in PSD order. In non-drifting eigendirections of* Q_ρ*,* $\mathcal{I}_{\min}$ *contributes zero; under full-rank drift, full-state detectability, and simultaneous diagonalizability,* $\mathcal{I}_{\min}$ *is positive definite.*

Theorem 4.2 asserts that survival is a *Fisher-information lower bound on the agent's policy*, not a designer's information budget — the agent must source enough information in the directions where drift acts to keep the steady-state covariance bounded. The associated matrix Lagrange multiplier $\Lambda \succeq 0$ produces the action-selection rule (22) in which the exploration bonus is the trace inner product

$\text{Tr}(\Lambda \cdot \mathcal{I}_o(a))$ — the matrix analog of the scalar bonus $\lambda_{\text{surv}} \cdot \text{CIY}(a)$. Three remarks unpack what the bound says.

Linear-Gaussian scope The matrix derivation assumes linear-Gaussian observation; outside this regime, FIM is no longer the exact information-yield rate and the information-form Kalman update's additivity (Appendix G.2) requires local linearization or extended-Kalman treatment. The qualitative dual-drive picture survives; the LMI form does not. This is the dominant scope restriction.

Pathwise vs. expectation gap; iid hypothesis Theorem 4.2's expectation form assumes iid action draws and controls the deterministic DARE; the iid bound lifts to a pathwise cumulative-FIM bound (Lemma A.1) but does *not* control the per-step covariance trajectory P_t when the support contains a near-degenerate-FIM action on the drifting subspace — iid bursts of length $\Theta(\log T)$ can drive $\lambda_{\max}(P_t) > R^2$ regardless of expectation slack. Closing this gap requires per-action structure: Theorem A.2 (full-space per-action LMI ⇒ deterministic pathwise survival, strongest hypothesis) or Theorem A.3 (drift-axis-projected per-action LMI under standard reduced-Kalman structural conditions ⇒ deterministic pathwise survival on each subspace, intermediate hypothesis), in Appendix A.6 — both deliver pathwise survival under *arbitrary* (not necessarily iid) action schedules. Markov-policy generalization of the expectation form is left open. The directional discrimination of Proposition 4.3 is robust to the iid/pathwise gap; Sinopoli et al. 2004 [46] flags the parallel expected-vs-pathwise distinction in scalar-SISO intermittent Kalman.

4.3 The blank-wall resolution

Theorem 4.2 gives a Fisher-information lower bound on the policy's averaged FIM, but doesn't by itself ensure that the agent's *exploration bonus* — the trace inner product with the matrix Lagrange multiplier Λ — credits actions only for information in directions that actually matter. The scalar analog of (6), the magnitude bound $\mathbb{E}_\pi[U_o(a)] \leq U_o^{\max}$, makes the failure mode visible: in an n-dimensional environment whose drift acts on a k-dimensional subspace $S_\rho \subset \mathbb{R}^n$, an action whose observation channel is aligned *orthogonal* to S_ρ — a "blank-wall" action — drives scalar $\mathbb{E}_\pi[U_o]$ arbitrarily small while contributing zero information about the drifting coordinates. The matrix lift closes this under a structural hypothesis on Λ:

Proposition 4.3 (Blank-wall receives zero survival bonus)**.** *Let a_w be a blank-wall action:* $\text{range}(\mathcal{I}_o(a_w)) \perp \text{range}(\mathcal{I}_{\text{min}})$*, and assume the controller's Λ satisfies* $\text{range}(\Lambda) \subseteq \text{range}(\mathcal{I}_{\text{min}})$ *(a controller-design hypothesis; satisfied by $\Lambda \propto \mathcal{I}_{\text{min}}$). Then* $\text{Tr}(\Lambda\, \mathcal{I}_o(a_w)) = 0$*, so a_w contributes zero matrix exploration bonus regardless of magnitude.*

Proposition 4.3 gives directional discrimination by structure: the matrix Lagrangian credits the agent only for information acquired in the drifting subspace, regardless of how much information is available off-drift. The full F-A-G-P enforcement story for the matrix family member $\Lambda \propto \mathcal{I}_{\text{min}}$, which satisfies the range-containment hypothesis by construction, is in Appendix A.7.

Anisotropic-drift dependence The $\Lambda \propto \mathcal{I}_{\text{min}}$ calibration is keyed to Q_ρ structure; under near-isotropic drift the argmax can flip to high-magnitude wall-aligned actions. The $\Lambda \propto \mathcal{I}_{\text{min}}^p$ refinement (Appendix A.6) recovers directional concentration with closed-form critical exponent p_{crit} diverging as $\sigma_x/\sigma_y \to 1$: the matrix-LMI value proposition rests on anisotropic drift.

4.4 Empirical anchor

A 2D Gaussian environment with one drifting axis (σ_w swept) and one stationary axis realizes the structure (full setup, action set, and hyperparameters in Appendix H.1). Three controllers differ only in the exploration bonus: greedy (0), scalar margin-triggered ($\propto \text{Tr}(\mathcal{I}_o(a))$), and LMI margin-triggered ($\text{Tr}(\Lambda \cdot \mathcal{I}_o(a))$ with $\Lambda \propto \mathcal{I}_{\text{min}}$). The action set includes a `wall_extreme` ε*-blank-wall* action with $\mathcal{I}_o = \text{diag}(0.0625, 25)$ — drift-axis FIM small but not exactly zero, stationary-axis FIM dominant; Proposition 4.3 covers the $\varepsilon = 0$ idealization, and the failure persists empirically at $\varepsilon = 0.0625$.

The pattern instantiates Proposition 4.3's failure mode (scalar controller dies via blank-wall capture; at $\sigma_w = 2.0$ it accumulates 383 units of Fisher information on the stationary axis and 0.96 on the drifting axis — two orders of magnitude in the wrong direction) and validates its resolution (LMI controller selects `drift_probe` 100% under directional $\Lambda \propto \mathcal{I}_{\text{min}}$). Full reading-by-reading discussion in

Table 1: Finite-horizon survival rate / mean reward across σ_w for the three controllers (100 episodes × 200 steps each, Model S Gaussian noise). At $\sigma_w \geq 1.5$ the LMI controller selects 'drift_probe' 100% and survives at zero exploit reward (drift-aligned probing dominates the Q-value calculation); greedy and scalar select 'wall_extreme' and die rapidly. At low drift, LMI matches greedy bit-identically — the survival drive is silent.

σ_w	greedy survival / reward	scalar survival / reward	LMI survival / reward
0.5	95% / 147.5	71% / 0	95% / 147.5
1.0	16% / 61.4	1% / 0	16% / 61.4
1.5	0% / 25.6	0% / 0	**100% / 0**
2.0	0% / 16.5	0% / 0	**100% / 0**
2.5	0% / 11.7	0% / 0	**100% / 0**
3.0	0% / 8.5	0% / 0	**100% / 0**

Appendix H. The experiment validates *directional discrimination and finite-horizon survival*, not a smooth survival/reward Pareto frontier — the action set's single drift-aligned probe drives covariance well below R^2, so the margin trigger transitions bistably rather than modulating dynamically (full discussion in Appendix B.5).

5 Mechanism

We narrate how the Lyapunov persistence inequality, the agent's Bayesian update gain, and the matrix Lagrange multiplier on the Fisher Information Matrix compose to produce a *directional* survival mandate. Full proofs of the four supporting results are in Appendix A (DARE feasibility for Theorem 4.2), Appendix A.5 (matrix-Chernoff pathwise lift), Appendix A.6 (per-action LMI strengthenings), and Appendix A.7 (F-A-G-P enforcement).

The key ideas decompose into four steps.

Step 1: Coupling the update gain to Lyapunov persistence Substituting the Bayesian-update bridge $\alpha = \eta^* c$ (scalar normalized observation, c the fixed channel sensitivity, Appendix F.1) into the persistence inequality (5) gives the *survival ceiling* $U_o^{\max}(M_t) = U_M(Rc/\rho - 1)$ — positive in the feasible regime $Rc > \rho$, vanishing linearly as $U_M \to 0$. The structural content is the proportionality: low model uncertainty *forces* a vanishing observation-noise budget, so any controller respecting the budget intensifies probing at low uncertainty rather than at high. The full KKT setup ((15)–(17)) and the survival-margin controller family construction are in Appendix F.2.

Step 2: The blank-wall failure of the scalar magnitude form The scalar constraint $\mathbb{E}_\pi[U_o(a)] \leq U_o^{\max}$ bounds *aggregate* observation noise — it cares about magnitude, not direction. In an n-dimensional environment where drift acts only on a k-dimensional subspace S_ρ, a pathological policy can satisfy the bound trivially: choose actions whose observation channel is aligned *orthogonal* to S_ρ. Such "blank-wall" actions drive scalar U_o arbitrarily close to zero while contributing no information about the drifting coordinates; the agent's mismatch in S_ρ accumulates without bound, and the agent dies while satisfying the math. This is the multidimensional analog of the bursting result of Anderson 1985 [9] under low persistent excitation: the agent has *some* identifying information but in the wrong directions.

Step 3: Matrix LMI lift via range-orthogonality The fix is to lift the constraint from scalar magnitude to matrix structure. The action-conditional Fisher Information Matrix $\mathcal{I}_o(a) = H^\top R_o(a)^{-1} H$ enters the information-form Kalman recursion additively, and the deterministic DARE under the *averaged* FIM admits a stabilizing steady-state solution Σ_δ with $\lambda_{\max}(\Sigma_\delta) < R^2$ when (6) holds — this is Theorem 4.2. The constrained policy program now carries a positive-semidefinite matrix Lagrange multiplier $\Lambda \succeq 0$, and the optimal action selection rule (22) replaces the scalar exploration bonus $\lambda_{\text{surv}} \cdot \text{CIY}(a)$ with the trace inner product $\text{Tr}(\Lambda \cdot \mathcal{I}_o(a))$ — the matrix analog. Under the controller-design hypothesis $\text{range}(\Lambda) \subseteq \text{range}(\mathcal{I}_{\min})$, satisfied by construction by the matrix family member $\Lambda \propto \mathcal{I}_{\min}$, range-orthogonality forces $\Lambda \mathcal{I}_o(a_w) = 0$ for any blank-wall action a_w supported orthogonally to the drifting subspace; the trace exploration bonus $\text{Tr}(\Lambda \cdot \mathcal{I}_o(a_w))$ vanishes. This is

Proposition 4.3: the agent's exploration bonus discriminates by *direction*, not by magnitude, so the failure mode of Step 2 is closed by structure.

Step 4: From expectation to pathwise Theorem 4.2 controls only the steady-state DARE under expected FIM, not per-step P_t under realized draws — iid bursts of a near-degenerate-FIM action can briefly violate $\lambda_{\max}(P_t) < R^2$. The operational pathwise condition is per-action: Theorem A.2 (full-space) and Theorem A.3 (drift-axis-projected, under reduced-Kalman structural conditions) both deliver deterministic pathwise survival (Appendix A.6); the matrix-Chernoff lemma Lemma A.1 gives a parallel high-probability bound on cumulative FIM.

In summary, the agent's policy faces a Fisher-information lower bound supported on the drifting subspace, derived from Lyapunov persistence rather than from priors or external rewards. The matrix Lagrange multiplier is the canonical mechanism that lifts this from a magnitude bound to a directional mandate. The "tragedy" the paper's title names is the *structural inversion* itself: in the standard contract the agent explores when uncertain and exploits when confident, but the persistence inequality coupled to the Bayesian update gain reverses this — exploration is forced *precisely at maximum confidence*, where the survival-noise budget collapses to zero. The static-program multiplier on the survival constraint is finite; what diverges is the worst-case adversarial exit-time clock from Lemma 3.1 (ii) — infinite above threshold, finite below. The divergent gain in Definition 4.1 is a controller-design overshoot of the finite multiplier, ensuring the controller's response loop closes inside that finite exit-time window. Divergence is on the environment-side clock, not the controller-side multiplier.

6 Limitations and Conclusion

6.1 Limitations

Three §4 caveats are not re-listed: linear-Gaussian scope, pathwise-vs-expectation gap, anisotropic-drift dependence (Appendix A.6).

CIY-U_o surrogate CIY $\propto 1/U_o$ is exact in 1D Gaussian-linear and heuristic outside; the matrix FIM derivation removes the surrogate (Appendix F.3).

Policy-lift Jensen gap $\mathbb{E}_\pi[U_o] \leq U_o^{\max}$ is a Jensen-strict sufficient condition for $\mathbb{E}_\pi[\mathcal{I}_o] \succeq \mathcal{I}_{\min}$; converse with $(1+\kappa)^2/(4\kappa)$ Kantorovich slack (Appendix G.6).

Bistable controller + Pareto frontier The 2D experiment's k_L gain governs a bistable survival-vs-reward Pareto frontier with no smooth tradeoff at this resolution; smoothing likely requires stochastic policies or richer action sets (Appendix B.5).

KKT shadow-price as LP multiplier; static-vs-multi-step The static-program multiplier on (15) is finite (LP slope) and DARE variance sensitivity is independently bounded (Appendix E); Definition 4.1's divergent gain is a *controller-design overshoot* justified environment-side by the worst-case exit time of Lemma 3.1 (ii), not a KKT-derived requirement. The multi-step CMDP companion to formalize the exit-time framing is left open. Full discussion in Appendix F.2 (cf.[10,19]).

Family-member enforcement; LMI sufficiency Proposition A.4 (Appendix A.7) recovers F-A-G-P enforcement: A (drift-aligned bonus) is trivial in 1D, non-trivial in multi-D — naïve $\lambda \cdot \mathrm{Tr}(\mathcal{I}_o)$ violates A and *is* the blank-wall failure mode; $\Lambda \propto \mathcal{I}_{\min}$ satisfies A by construction. Theorem 4.2 is sufficient, not necessary (Appendix A.4); multi-agent settings are out of scope.

6.2 Conclusion

A confident agent in a drifting world is not safe. The Lyapunov inequality governing whether correction outpaces drift induces a structural exploration mandate at low uncertainty — a directional Fisher-information lower bound on the agent's policy, with range-containment of Λ denying blank-wall actions any LMI reward. The result is a structural (Lyapunov-persistence) obstruction to the dark-room failure mode *in drifting tracking environments*, composing four lineages — adaptive control under drift, dual control, directional FIM experiment design, and the post-2020 dark-room debate — that have lived separately for four decades.

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A DARE-Imported $\mathcal{I}_{\text{min}}$ and Sufficiency Restrictions

The closed-form expression for the survival floor $\mathcal{I}_{\text{min}}(Q_\rho, A, R^2)$ is theorem-imported standard control theory. We record it here for completeness, along with the precise restriction under which Theorem 4.2 admits a converse.

A.1 DARE form

For the linear-Gaussian observation model (21) with state-transition matrix A, process noise covariance Q_ρ, observation matrix H, and steady-state observation noise covariance R_o, the discrete-time algebraic Riccati equation (DARE) for the steady-state covariance Σ_δ is

$$\Sigma_\delta = A\Sigma_\delta A^\top + Q_\rho - A\Sigma_\delta H^\top (H\Sigma_\delta H^\top + R_o)^{-1} H\Sigma_\delta A^\top. \tag{7}$$

Under detectability of (A, H) and stabilizability of $(A, Q_\rho^{1/2})$, (7) admits a unique positive-semidefinite stabilizing solution Σ_δ^*, and $\lambda_{\text{max}}(\Sigma_\delta^*) < R^2$ is achieved if the action policy provides FIM $\mathbb{E}[\mathcal{I}_o(a)] = H^\top R_o^{-1} H$ dominating the unstabilizable modes of A scaled by Q_ρ/R^2. Stabilizability is to be read in the generalized sense allowing A-marginally-stable modes (e.g., $\lambda(A) = 1$ in DT, $\operatorname{Re}\lambda(A) = 0$ in CT) that are uncontrollable from $Q_\rho^{1/2}$ but observable from H — a relaxation that suffices for DARE existence and is the case relevant to the Appendix H setup where $A = I$ and $Q_\rho = \operatorname{diag}(\sigma_w^2, 0)$ make the y-axis A-marginally-stable and uncontrollable-from-$Q_\rho^{1/2}$, with observation feedback through H supplying the closed-loop contraction. The explicit lower bound $\mathcal{I}_{\text{min}} \succeq 0$ is obtained by solving (7) for the maximum Σ_δ consistent with $\lambda_{\text{max}}(\Sigma_\delta) < R^2$ and inverting; the closed form follows the standard treatment in[45] §4.4. $\mathcal{I}_{\text{min}}$ contributes zero in non-drifting eigendirections of Q_ρ; under full-rank drift it is positive definite.

A.2 Proof of Lemma 3.1 (persistence threshold)

We restate the lemma for convenience and prove the three parts. Notation: $V(\delta) := \frac{1}{2}\|\delta\|^2$, $\mathcal{B}_R := \{\delta : \|\delta\| \le R\}$.

Part (i) — sufficient direction (forward, all sector-bounded F) Under the sector condition (3), for any $\delta \in \mathcal{B}_R$ and any w with $\|w\| \le \rho$, $\dot V = \delta^\top \dot\delta = \delta^\top(-F(\delta) + w) \le -\alpha\|\delta\|^2 + \rho\|\delta\|$, where the second step uses (3) for the $-F$ term and Cauchy–Schwarz for the w term. The right side is strictly negative for $\|\delta\| > \rho/\alpha$. If $\alpha > \rho/R$ then $\rho/\alpha < R$, so on the boundary $\|\delta\| = R$ we have $\dot V < 0$; the sublevel set $\{V \le R^2/2\} = \mathcal{B}_R$ is therefore positively invariant. Ultimate boundedness with $R^* = \rho/\alpha$ follows from the standard sector-Lyapunov ultimate-boundedness theorem[43] Theorem 4.18. □

Part (ii) — worst-case sharp (reverse direction, finite exit-time witness) Suppose F satisfies the upper sector bound $\delta^\top F(\delta) \le (\alpha+\gamma)\|\delta\|^2$ on $\mathcal{B}_R$ for some $\gamma \ge 0$, and choose the adversarial disturbance $w^*(t) = \rho\,\delta(t)/\|\delta(t)\|$ (admissible since $\|w^*\| = \rho$, defined wherever $\delta \ne 0$). Then $\delta^\top w^* = \rho\|\delta\|$, so $\frac{d}{dt}\|\delta\|^2 = 2\delta^\top\dot\delta \ge -2(\alpha+\gamma)\|\delta\|^2 + 2\rho\|\delta\|$, which yields $d\|\delta\|/dt \ge -(\alpha+\gamma)\|\delta\| + \rho$ wherever $\delta \ne 0$. Compare to the linear ODE $\dot\mu = -(\alpha+\gamma)\mu + \rho$ with $\mu(0) = \|\delta_0\|$: by the comparison principle[43] Lemma 3.4, $\|\delta(t)\| \ge \mu(t)$ for all t. The linear ODE has closed-form solution $\mu(t) = \frac{\rho}{\alpha+\gamma} + \left(\|\delta_0\| - \frac{\rho}{\alpha+\gamma}\right)e^{-(\alpha+\gamma)t}$, which converges *upward* to $\rho/(\alpha+\gamma)$ when $\|\delta_0\| < \rho/(\alpha+\gamma)$. If $\rho/(\alpha+\gamma) > R$ — equivalently $\alpha+\gamma < \rho/R$ — then $\mu(t)$ crosses R in finite time. Solving $\mu(T) = R$ gives the exit-time bound (4). Since $\|\delta(t)\| \ge \mu(t)$, the trajectory leaves $\mathcal{B}_R$ no later than the time at which μ does. □

Part (iii) — tight-sector / universal-trajectory iff Setting $\gamma = 0$ in (ii) — which corresponds to the linear correction $F(\delta) = M\delta$ saturating the upper sector bound at the lower sector bound — combines with (i) to give: $\alpha > \rho/R \iff$ all admissible disturbances fail to drive the agent out of $\mathcal{B}_R$. The sufficiency direction is (i). The necessity direction: if $\alpha \le \rho/R$, part (ii) at $\gamma = 0$ gives w^* that exits $\mathcal{B}_R$ in finite time bounded by (4) at $\gamma = 0$. Hence both implications hold and the threshold inequality (5) is two-sided. □

A.3 Scalar pedagogical limit

In the scalar random-walk limit ($A = 1$, scalar $H = 1$, scalar $Q_\rho = q$, scalar capacity R^2), the DARE (7) reduces to a quadratic in the steady-state variance P: $P = P + q - \frac{P^2}{P+R_o}$, giving $R_o = P^2/q - P$ at the boundary. Setting $P = R^2$ (boundary capacity), the corresponding observation-noise threshold is $R_o^{\max} = R^2(R^2 - q)/q = R^2(R^2/q - 1)$, valid when $q < R^2$. The corresponding scalar Fisher-information floor is

$$\mathcal{I}_{\min}^{\text{scalar}} = \frac{1}{R_o^{\max}} = \frac{q}{R^2(R^2 - q)}. \tag{8}$$

In the further limit where the agent's mismatch dynamics are coupled to its Kalman gain (as in Appendix F), the connection $q \leftrightarrow \rho$ is via the per-step drift variance under Model S, and the threshold (8) recovers the Appendix F form modulo the surrogate map between U_o and CIY.

A.4 Sufficiency vs necessity

Theorem 4.2's sufficient condition (6) becomes necessary under additional structure. The simplest sufficient further condition is *simultaneous diagonalizability*: the family $\{\mathcal{I}_o(a)\}_{a\in\mathcal{A}}$ commutes pairwise and commutes with Q_ρ, so that the LMI block-diagonalizes. In that case the LMI (6) decouples into per-eigendirection scalar inequalities, each of which is necessary and sufficient by the scalar DARE; the converse direction is constructively witnessed in each diagonal eigendirection k by the worst-case adversarial disturbance $w_k^*(t) = \rho_k\, \delta_k(t)/|\delta_k(t)|$, which is the matrix lift of Lemma 3.1 (iii). Under arbitrary action schedules without commutation, the per-step Riccati update is nonlinear in $\mathcal{I}_o(a_t)$ and expectation alone does not control pathwise covariance; (6) remains sufficient under stationary i.i.d. randomized policies but a converse requires the structural restriction.

A.5 Pathwise cumulative-FIM lower bound (proof of Lemma 4.1.1)

We restate Lemma 4.1.1 and supply the proof.

Lemma A.1 (Pathwise cumulative-FIM lower bound)**.** *Under iid sampling $a_1, \ldots, a_T \sim \pi$ with bounded FIM $0 \preceq \mathcal{I}_o(a) \preceq L \cdot I_d$ a.s. and $\mathbb{E}_\pi[\mathcal{I}_o(a)] \succeq \mathcal{I}_{\min}$ where $\mathcal{I}_{\min}$ is positive semidefinite with rank $k := \dim(\mathrm{range}(\mathcal{I}_{\min}))$ supported on the drifting subspace $\Pi_\rho := \mathrm{range}(\mathcal{I}_{\min})$, and $\iota_{\min} := \lambda_{\min}(\Pi_\rho \mathcal{I}_{\min} \Pi_\rho|_{\mathrm{range}(\Pi_\rho)}) > 0$, then for every $\delta \in (0,1)$, with probability at least $1-\delta$,*

$$\lambda_{\min}\left(\Pi_\rho \sum_{t=1}^{T} \mathcal{I}_o(a_t)\, \Pi_\rho\Big|_{\mathrm{range}(\Pi_\rho)}\right) \geq T\iota_{\min} - \sqrt{2LT\iota_{\min}\log(k/\delta)}. \tag{9}$$

(Full-rank case: $k = d$ and $\Pi_\rho = I$, recovering the unprojected statement.)

Proof. Apply the matrix Chernoff lower-tail bound [47] Theorem 5.1.1. For independent random Hermitian matrices $\{X_t\}$ with $0 \preceq X_t \preceq L \cdot I$ a.s., setting $Y = \sum_t X_t$ and $\mu_{\min} = \lambda_{\min}(\mathbb{E}[Y])$, Tropp's inequality (5.1.5) gives, for $t \in [0,1)$, $\Pr\big[\lambda_{\min}(Y) \leq t\,\mu_{\min}\big] \leq d \cdot \exp\Big(-\frac{(1-t)^2 \mu_{\min}}{2L}\Big)$. Set $X_t = \mathcal{I}_o(a_t)$ acting on Π_ρ (so $d \to k$ on the projected subspace). Then $\mu_{\min} = T\lambda_{\min}(\Pi_\rho \mathbb{E}_\pi[\mathcal{I}_o(a)]\Pi_\rho) \geq T\iota_{\min}$. Setting the right-hand side equal to δ and solving for $t\mu_{\min}$ gives $t\mu_{\min} = \mu_{\min} - \sqrt{2L\mu_{\min}\log(k/\delta)}$. The map $x \mapsto x - \sqrt{2Lx\log(k/\delta)}$ is monotone increasing for $x > L\log(k/\delta)/2$; in the meaningful regime $T\iota_{\min} \geq 2L\log(k/\delta)$ where the RHS is positive (the bound is vacuous outside this regime), substituting $\mu_{\min} \geq T\iota_{\min}$ and rearranging yields the claim. □

The lemma uses no commutativity hypothesis on $\{\mathcal{I}_o(a)\}$; matrix Chernoff is sharper than matrix Bernstein for PSD summands and gives the right $\sqrt{T}$ scaling for our application. The bound matches the rate at which a Lyapunov-survival argument needs to absorb $T \cdot Q$ worth of accumulated drift uncertainty: cumulative drift growth is linear in T and cumulative information is linear in T minus a $\sqrt{T}$ correction, so the information-acquisition rate dominates the drift-uncertainty growth rate with high probability for large T. The lemma controls cumulative information, not the per-step Riccati state P_t; pathwise control of P_t requires the additional per-action structure discussed in Appendix G.3 and Appendix A.6.

A.6 Operational safety factor γ and anisotropic $\Lambda \propto \mathcal{I}_{\text{min}}^p$

Theorem 4.2's expectation form is the *weakest* member of an LMI-strength gradient. The strongest is *per-action* LMI on the policy support, which gives deterministic pathwise survival.

Theorem A.2 (Per-action LMI $\Rightarrow$ deterministic pathwise survival)**.** *If every action in the policy support satisfies $\mathcal{I}_o(a) \succeq \mathcal{I}_{\text{min}}$, and the deterministic DARE under $\mathcal{I}_o \equiv \mathcal{I}_{\text{min}}$ has solution $\bar{P}$ with $\lambda_{\text{max}}(\bar{P}) < R^2$, then for any action schedule $\{a_t\}$ (not necessarily iid or stationary) and any $P_0 \preceq \bar{P}$, the Riccati recursion satisfies $\lambda_{\text{max}}(P_t) \le \lambda_{\text{max}}(\bar{P}) < R^2$ for all $t \ge 0$. For general P_0, the conclusion holds for all $t \ge T^*$ where T^* depends on P_0 and the recursion's contraction rate.*

Proof. Show by induction on t that $P_t \preceq P_t^{\text{const}}$ in the PSD order, where P_t^{const} is the trajectory under constant data $\mathcal{I}_{\text{min}}$ from the same P_0. *Base case:* $P_0 = P_0^{\text{const}}$ trivially. *Update step* (information form): $J_{t|t} = J_{t|t-1} + \mathcal{I}_o(a_t)$ where $J = P^{-1}$. Under $\mathcal{I}_o(a_t) \succeq \mathcal{I}_{\text{min}}$ and the inductive hypothesis $J_{t|t-1} \succeq J_{t|t-1}^{\text{const}}$, we get $J_{t|t} \succeq J_{t|t-1}^{\text{const}} + \mathcal{I}_{\text{min}} = J_{t|t}^{\text{const}}$, hence $P_{t|t} \preceq P_{t|t}^{\text{const}}$ (matrix inversion is order-reversing on PSD). *Predict step:* $P_{t+1|t} = AP_{t|t}A^\top + Q_\rho$, monotone in $P_{t|t}$ under fixed A, Q_ρ, so the order is preserved. Under $P_0 \preceq \bar{P}$, monotonicity of the constant-$\mathcal{I}_{\text{min}}$ recursion[45] §4.4 gives $P_t^{\text{const}} \preceq \bar{P}$ for all t. Therefore $\lambda_{\text{max}}(P_t) \le \lambda_{\text{max}}(P_t^{\text{const}}) \le \lambda_{\text{max}}(\bar{P}) < R^2$. The general-$P_0$ statement follows from finite-time contraction of the Riccati recursion to a neighborhood of $\bar{P}$. $\square$

Theorem A.3 (Drift-axis-projected per-action LMI $\Rightarrow$ deterministic pathwise survival)**.** *Suppose - **(i)** $\Pi_\rho \mathcal{I}_o(a) \Pi_\rho \succeq \iota_{\text{min}}^* \Pi_\rho$ for every $a \in \operatorname{supp}(\pi)$, with $\iota_{\text{min}}^* > 0$ satisfying the per-axis DARE on Π_ρ with $\bar{P}_{\iota_{\text{min}}^*} \prec R^2 I_{\Pi_\rho}$; - **(S1)** $A\Pi_\rho = \Pi_\rho A$ (A-invariance of the drift subspace); - **(S2)** $Q_\rho \Pi_\rho^\perp = 0$ (process noise supported on the drift subspace, automatic under $\Pi_\rho := \operatorname{range}(Q_\rho)$); - **(S3)** $\Pi_\rho \mathcal{I}_o(a) \Pi_\rho^\perp = 0$ for every $a \in \operatorname{supp}(\pi)$ (block-diagonal observation FIM, automatic under simultaneous-diagonalizability of $\{\mathcal{I}_o(a)\}_{a \in \operatorname{supp}(\pi)}$ with Q_ρ); - **(S4)** $\|\Pi_\rho^\perp A \Pi_\rho^\perp\|_2 \le 1$ (orthogonal A-block non-expansive); - **(S5)** initial covariance block-diagonal: $\Pi_\rho P_0 \Pi_\rho^\perp = 0$, with $\Pi_\rho P_0 \Pi_\rho \preceq \bar{P}_{\iota_{\text{min}}^*}$ on Π_ρ and $\|\Pi_\rho^\perp P_0 \Pi_\rho^\perp\|_2 < R^2$.*

Then for any *action schedule* $\{a_t\} \subseteq \operatorname{supp}(\pi)$ *— not necessarily iid or stationary —* deterministically*: $\lambda_{\text{max}}(P_t) < R^2$ for all $t \ge 0$.*

Proof. Hypotheses (S1)–(S3) and (S5) together imply the covariance recursion is *block-decoupled* w.r.t. $(\Pi_\rho, \Pi_\rho^\perp)$ at every step. *Update* (information form): $J_{t|t} = J_{t|t-1} + \mathcal{I}_o(a_t)$, with both summands block-diagonal by (S3) and the running J_t block-diagonal by induction from (S5); equivalently $(P_{t|t})_{11} = ((P_{t|t-1})_{11}^{-1} + \Pi_\rho \mathcal{I}_o(a_t) \Pi_\rho)^{-1}$ and similarly for the (2,2) block, with no cross-coupling. *Predict*: $P_{t+1|t} = AP_{t|t}A^\top + Q_\rho$, with A block-diagonal by (S1) and Q_ρ supported on Π_ρ by (S2); block-diagonal $P_{t|t}$ propagates to block-diagonal $P_{t+1|t}$ as $(P_{t+1|t})_{11} = A^{\rho\rho}(P_{t|t})_{11}(A^{\rho\rho})^\top + Q_\rho|_{\Pi_\rho}$ and $(P_{t+1|t})_{22} = A^{\perp\perp}(P_{t|t})_{22}(A^{\perp\perp})^\top$.

The (1,1) block runs a *closed* sub-Riccati on Π_ρ with per-step data lower-bounded by $\iota_{\text{min}}^* I_{\Pi_\rho}$ (hypothesis (i)). Apply Theorem A.2 on the sub-system: $(P_{t|t})_{11} \preceq \bar{P}_{\iota_{\text{min}}^*}$ for all $t \ge 0$ given $(P_0)_{11} \preceq \bar{P}_{\iota_{\text{min}}^*}$ (from (S5)). Hence $\lambda_{\text{max}}((P_t)_{11}) \le \lambda_{\text{max}}(\bar{P}_{\iota_{\text{min}}^*}) < R^2$.

The (2,2) block sees no process noise injection (by (S2)): predict scales by $A^{\perp\perp} \otimes A^{\perp\perp}$, update is non-increasing in covariance (additive PSD precision). By (S4), $\|(P_{t+1|t})_{22}\|_2 \le \|A^{\perp\perp}\|_2^2 \cdot \|(P_{t|t})_{22}\|_2 \le \|(P_{t|t})_{22}\|_2$. So $\|(P_t)_{22}\|_2$ is non-increasing in t and bounded by $\|(P_0)_{22}\|_2 < R^2$ (from (S5)).

Block-diagonal P_t gives $\lambda_{\text{max}}(P_t) = \max(\lambda_{\text{max}}((P_t)_{11}), \lambda_{\text{max}}((P_t)_{22})) < R^2$ for all $t \ge 0$. $\square$

Theorem A.3 fills the gap between Theorem 4.2's expectation form (weakest end of the LMI-strength gradient — sufficient deterministically on the DARE solution, sufficient pathwise on cumulative FIM) and Theorem A.2's full-space per-action form (strongest end — deterministic pathwise survival under arbitrary action schedules). Drift-axis-projected per-action LMI is the natural intermediate: it relaxes Theorem A.2's full-space hypothesis (often too strong — most natural action sets contain wall-aligned actions with zero drift-axis FIM) while strengthening Theorem 4.2's expectation hypothesis to a per-action condition on the drifting subspace, with the same deterministic conclusion as the strong end. Hypotheses (S1)–(S5) are standard reduced-Kalman structure: (S1)/(S2) are the drift-subspace decomposition itself, (S3) is the simultaneous-diagonalizability already invoked for closed-form $\mathcal{I}_{\text{min}}$ in Appendix A.3, (S4) is non-expansiveness of the orthogonal A-block (automatic under $A = I$), and (S5) is initial-uninformative covariance — all satisfied in the Appendix H setup.

Operational safety factor Define the strict feasibility set with safety factor $\gamma > 1$:

$$\mathcal{A}_F^{\text{op}}(\gamma) := \{a : \mathcal{I}_o(a) \succeq \gamma \cdot \mathcal{I}_{\text{min}}\}.$$

A controller whose support sits inside $\mathcal{A}_F^{\text{op}}(\gamma)$ has *operational margin* γ on the per-step LMI. The Appendix H LMI controller's support is a singleton (`drift_probe` at high drift, `exploit` at low drift) with operational margin $\gamma \in [50, 960]$ across $\sigma_w \in [1.0, 3.0]$; per Theorem A.2 the *covariance recursion* is deterministically bounded with $\lambda_{\max}(P_t) \le \lambda_{\max}(\bar{P}) < R^2$ throughout, the action schedule is a deterministic argmax, and the observed 100/100 finite-horizon survival under Model S Gaussian noise is the natural manifestation of these two deterministic facts on a covariance margin slack of $1 - \lambda_{\max}(\bar{P})/R^2 \in [50/51,\ 960/961]$. The greedy controller's per-action margin slides from $\gamma_{\text{greedy}} = 112$ at $\sigma_w = 0.5$ to $\gamma_{\text{greedy}} = 1.4$ at $\sigma_w = 3.0$, tracing the gradient between Theorem 4.2's weak end and Theorem A.2's strong end.

Anisotropic-drift refinement: $\Lambda \propto \mathcal{I}_{\text{min}}^p$. Under symmetric drift on multiple axes ($\sigma_x \approx \sigma_y$), the calibration $\Lambda \propto \mathcal{I}_{\text{min}}$ "leaks" weight onto small-magnitude eigendirections, allowing high-magnitude wall-aligned actions to satisfy per-step LMI but lack pathwise margin. Replacing $\Lambda \propto \mathcal{I}_{\text{min}}$ with $\Lambda \propto \mathcal{I}_{\text{min}}^p$ for $p > 1$ (with k rescaled to keep $\|\Lambda\|$ comparable across p) concentrates the bonus signal on the dominant drift direction. The bonus comparison `drift_probe` vs `wall_extreme` flips iff

$$(\mathcal{I}_{\text{min},0}/\mathcal{I}_{\text{min},1})^p \ > \ \Delta_w/\Delta_d,$$

where Δ_d, Δ_w are the per-action FIM ratios of the two candidate actions. This gives a closed-form critical exponent

$$p_{\text{crit}}(\sigma_x, \sigma_y) \ = \ \frac{\log(\Delta_w/\Delta_d)}{\log(\mathcal{I}_{\text{min},0}/\mathcal{I}_{\text{min},1})},$$

which matches the empirical breakdown across all four anisotropic (σ_x, σ_y) cells we tested (Appendix C.6). As $\sigma_x/\sigma_y \to 1$ (isotropic drift), $p_{\text{crit}} \to \infty$: no concentration of Λ along $\mathcal{I}_{\text{min}}$'s eigenstructure can recover directional discrimination, and the matrix-LMI value proposition rests on anisotropic drift.

A.7 Family-member enforcement: F-A-G-P hypotheses and Proposition A.6

We formalize the conditions under which a family member from Definition 4.1 actually selects a feasible action. The setup is the matrix-lift formulation of Appendix G (which specializes cleanly to 1D); $\mathcal{A}_F = \{a : \mathcal{I}_o(a) \succeq \mathcal{I}_{\text{min}}\}$ in the multi-D form, $\{a : U_o(a) \le U_o^{\max}(M_t)\}$ in the 1D form.

(F) Strict feasibility There exists $a^\dagger \in \mathcal{A}$ with $\Pi_M \mathcal{I}_o(a^\dagger) \Pi_M \succ \Pi_M \mathcal{I}_{\text{min}} \Pi_M$ on the drift subspace and $\mathcal{I}_o(a^\dagger) \succeq \mathcal{I}_{\text{min}}$ overall. (1D form: $a^\dagger$ exists with $U_o(a^\dagger) < U_o^{\max}$.)

(A) Drift-aligned bonus The matrix multiplier satisfies $\Lambda(M_t) = \Pi_M \Lambda(M_t) \Pi_M$, equivalently $\text{range}(\Lambda) \subseteq \text{range}(\mathcal{I}_{\text{min}})$. (Trivial in 1D.)

(G) Gain dominance Define the bonus separation

$$\Delta_g(\Lambda) := \min_{a^\dagger \in \mathcal{A}_F} \text{Tr}\big(\Lambda \mathcal{I}_o(a^\dagger)\big) - \max_{a \notin \mathcal{A}_F} \text{Tr}\big(\Lambda \mathcal{I}_o(a)\big).$$

There exists $\Delta_0 > 0$ such that $\Delta_g(\Lambda(M_t)) > G_{\max} := \sup_{a,a'} |Q_O(a) - Q_O(a')|$ for every M_t in the stress region $\{M : \Delta(M) < \Delta_0\}$.

(P) Argmax expressibility The policy class Π contains every Dirac policy δ_a for $a \in \mathcal{A}$.

Proposition A.4 (Family-member enforcement)**.** *Under F, A, G, P, the controller's argmax $a^*(M_t) \in \arg\max_a [Q_O(a) + \text{Tr}(\Lambda(M_t) \mathcal{I}_o(a))]$ lies in $\mathcal{A}_F$ for every M_t in the stress region; in particular, in the 1D specialization the controller enforces threshold* (14).

Proof. For any $a \notin \mathcal{A}_F$, by (G) and (A),

$$\mathrm{Tr}\big(\Lambda \mathcal{I}_o(a^\dagger)\big) - \mathrm{Tr}\big(\Lambda \mathcal{I}_o(a)\big) \;\geq\; \Delta_g(\Lambda) \;>\; G_{\max} \;\geq\; Q_O(a) - Q_O(a^\dagger),$$

so $Q_O(a^\dagger) + \mathrm{Tr}(\Lambda \mathcal{I}_o(a^\dagger)) > Q_O(a) + \mathrm{Tr}(\Lambda \mathcal{I}_o(a))$. Therefore $a^*(M_t) \in \mathcal{A}_F$, since otherwise a^* would be infeasible and would have score strictly less than $a^\dagger$, contradicting the argmax definition. □

Sharpness Each hypothesis is essential:

- **(F) violation:** $\mathcal{A}_F = \emptyset$ — the LMI is infeasible; no controller succeeds.
- **(A) violation:** Λ has range outside $\mathcal{I}_{\text{min}}$'s — wall-aligned actions can dominate the bonus and the Appendix G.1 blank-wall attack succeeds. Empirical witness: Appendix H at $\sigma_w \geq 2.83$, scalar-trace argmax `wall_extreme` is per-step LMI-infeasible.
- **(G) violation:** Λ grows too slowly (e.g., $\Lambda \equiv 0$ — pure greedy) — bonus separation stays below $G_{\max}$ and exploit wins outside $\mathcal{A}_F$. Empirical witness: greedy controller in Appendix H at high σ_w.
- **(P) violation:** policy class restricted away from argmax — degenerate.

1D vs multi-D In 1D, A is trivial (range is automatic) and G is implied by F-strict + family membership: under strict F (witness $a^\dagger$ with $U_o(a^\dagger) < U_o^{\max}$), $\Delta_g(\lambda) \geq \lambda \cdot c_+$ where $c_+ := 1/U_o(a^\dagger) - 1/U_o^{\max} > 0$, so $\lambda \to \infty$ at margin closure drives $\Delta_g \to \infty$; F + P thus suffice for either family member to enforce threshold (14). In multi-D, A is non-trivial: naïve scalar lifts $\lambda \cdot \mathrm{Tr}(\mathcal{I}_o(a))$ violate A, supplying the Appendix G.1 blank-wall failure mode; the matrix lift $\Lambda \propto \mathcal{I}_{\text{min}}$ used in Appendix G satisfies A by construction.

B Empirical Reproducibility

B.1 2D experiment hyperparameters (Appendix H)

Table 2: 2D experiment hyperparameters.

Parameter	Value	Description
R	4.0	Structural capacity (per-axis absolute mismatch ceiling)
σ_w	$\{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$	Per-step drift stddev (axis 0)
Q_ρ	$\mathrm{diag}(\sigma_w^2, 0)$	Process noise covariance
Action set	5 diagonal $R_o(a)$	See Table 8 in Appendix H.1
Q_O on exploit	1.0	Constant pragmatic reward
k_s (scalar gain)	1.0	Scalar margin-triggered
k_L (LMI gain)	500.0	LMI margin-triggered
Episodes per condition	100	Per controller per σ_w
Steps per episode	200	Maximum per episode
Initial P_0	I	Initial agent covariance

B.2 Per-axis FIM accumulation (highlighted finding)

At $\sigma_w = 2.0$, mean accumulated Fisher information per episode by axis (averaged over 100 episodes; surviving + dead episodes both included):

Table 3: Per-axis Fisher information accumulation at $\sigma_w = 2.0$.

Controller	Drift-axis FIM	Stationary-axis FIM
greedy	2.89	2.89
scalar	0.96	383.25
LMI	800.00	50.00

The scalar controller has accumulated 400x as much information on the stationary axis as on the drifting axis — the precise quantitative footprint of the blank-wall attack. The LMI controller, by contrast, has accumulated 16x as much on the drifting axis; the residual stationary-axis FIM comes from `drift_probe`'s nonzero $\mathcal{I}_o[1, 1] = 0.25$ per step, not from any active wall-probing.

B.3 1D legacy simulation hyperparameters (supplementary)

The 1D simulation (predates and does not test the Appendix G claims; included for the threshold-bite consistency check of Appendix H.5; not packaged in the current `sim/` directory but described here for reproducibility):

Table 4: 1D legacy simulation hyperparameters.

Parameter	Value	Description
R	4.0	Structural capacity
ρ	0.5	Mean per-step drift (Model D component)
σ_w	0.5	Per-step drift stddev
U_o^{exploit}	100.0	Observation noise under exploit
U_o^{explore}	0.01	Observation noise under explore
Initial U_M	1.0	Initial posterior variance
Episodes / steps	$300/200$	Per strategy
Reward	$\max(0, 1 - \lvert\delta_t\rvert)$	Awarded on exploit actions only

B.4 Random seeds and reproducibility

The 2D simulation uses `numpy.random.default_rng(seed)` with seed $\in \{0, 1, \ldots, 99\}$ for each controller × σ_w combination. Each seed initializes both the environment process noise and the agent observation noise; results are deterministic given the seed. Code: `sim/variant_causal_ib_2d.py` (single file, ~30 seconds wall-clock for the full sweep on a single CPU). Numerical results: `sim/results.json`, `sim/bistability.json`. Plots: `sim/plots/fig{1..5}_*.png`.

B.5 Bistable controller artifact

Both the 2D experiment and the 1D legacy experiment exhibit a sharp bistable transition between "always exploit" and "always probe" as the controller gain k crosses a threshold. In 1D the artifact derives from the binary action geometry: with $\text{CIY}(\text{exploit}) = 0.01$ and $\text{CIY}(\text{explore}) = 100$, the score comparison $Q_O(\text{exploit}) + \lambda \cdot 0.01$ vs $0 + \lambda \cdot 100$ has a single λ^* crossover; below it the agent always exploits, above it always probes. In 2D the artifact derives from the action authority: a single `drift_probe` step drives $P[0, 0]$ to ~0.2, which keeps the persistence margin $R^2 - \lambda_{\max}(P) \approx 16$ permanently saturated, so the matrix margin-triggered Λ does not modulate dynamically. In both cases, the implication is the same: smooth survival-vs-reward Pareto behavior is not achieved at this resolution, and the two controllers' policies become deterministic step-functions in the gain. Smoothing the tradeoff likely requires stochastic policies, action sets where individual probes do not dominate steady-state covariance, or alternative trigger functional forms; we leave this as open follow-up work.

C 2D Experiment Details and Open Variants

The Appendix H simulation runs a Model S 2D environment $\Omega^{(0)}_{t+1} = \Omega^{(0)}_t + w_t$ ($w_t \sim \mathcal{N}(0, \sigma_w^2)$), $\Omega^{(1)}_{t+1} = \Omega^{(1)}_t$ with $R = 4$, sweeping $\sigma_w \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$ at 100 episodes of 200 steps each per controller.

C.1 Action set rationale

The five actions have diagonal $R_o(a)$ chosen so that (i) `wall_extreme` has substantially higher $\text{Tr}(\mathcal{I}_o)$ than `drift_probe` (creating the trap a scalar controller falls into), (ii) `drift_probe` has high information specifically on the drifting axis (the action the LMI should pick), (iii) `exploit` provides nontrivial residual information on both axes, and (iv) all entries are diagonal in the same basis as Q_ρ (simultaneous diagonalizability). The simultaneous-diagonalizability assumption is the regime in which the LMI cleanly decouples into per-eigendirection scalar inequalities (Appendix A.4); off-diagonal R_o would test the more general LMI form.

C.2 Matrix margin-triggered Λ

The LMI controller uses $\Lambda = k_L \cdot \frac{\mathcal{I}_{\min}}{R^2 - \lambda_{\max}(P_t)}$ where $\mathcal{I}_{\min}$ is the per-axis DARE-derived floor (Appendix A.3). This is the matrix lift of the scalar margin-triggered family member (Definition 4.1): the scalar margin denominator is preserved, but the numerator's eigenstructure is set by the drift covariance via $\mathcal{I}_{\min}$. The range-containment range$(\Lambda) \subseteq$ range$(\mathcal{I}_{\min})$ holds by construction, satisfying the hypothesis of Proposition 4.3.

C.3 What the experiment establishes

(i) The blank-wall attack is reproducible in 2D: the scalar controller picks `wall_extreme` 100% of the time and accumulates ~100x more Fisher information on the stationary axis than the drifting axis. (ii) The LMI controller's directional structure prevents the trap: it picks `drift_probe` 100% of the time and never picks any wall-aligned action. (iii) At low drift, the LMI controller correctly matches the greedy controller — the survival drive's drift-specificity is empirically realized.

C.4 What the experiment does not establish

(i) A smooth survival-vs-reward Pareto frontier is not achieved; the matrix margin-triggered controller in this action geometry has a sharp bistable transition in k_L (jumping from 0% to 100% survival between $k_L = 300$ and $k_L = 400$ at $\sigma_w = 1.5$). The same artifact appears in the 1D analog (Appendix B). (ii) The action set is simultaneously diagonalizable with Q_ρ, so the LMI's full-generality directional structure is not stress-tested. (iii) $\mathcal{I}_{\min}$ is the steady-state DARE form; transient behavior of the bound is not characterized.

C.5 Stochastic-softmax extension

Replacing the LMI controller's deterministic argmax with softmax-randomized variants at temperatures $T \in \{0.1, 0.5, 1.0, 2.0\}$ tests whether directional discrimination survives policy randomization. At $T = 0.1$, the stochastic LMI is near-deterministic on `drift_probe` at high drift and achieves 100% survival, matching the deterministic controller's behavior. As T increases, action mass spreads across `drift_probe` / `wall_probe` / `diag_probe`, and survival drops sharply at high drift. Selected results at $\sigma_w \in \{1.0, 2.0, 2.5\}$ (full sweep in supplementary materials):

Table 5: Stochastic-softmax LMI: survival rate / drift-axis FIM accumulated, across temperatures.

σ_w	$T = 0.1$	$T = 0.5$	$T = 1.0$	$T = 2.0$
1.0	24% / 19	62% / 170	72% / 186	72% / 187
2.0	100% / 800	90% / 709	42% / 363	14% / 165
2.5	100% / 800	100% / 800	86% / 716	30% / 347

(Survival rate / drift-axis FIM accumulated.) The directional-discrimination mechanism survives in the low-temperature regime where the policy is approximately deterministic; it cracks sharply at high temperature where the policy averages information across all axes — at $\sigma_w = 2.0$, raising T from 0.5 to 2.0 collapses survival from 90% to 14%.

Across the 20 stochastic-softmax variants, the policy's *Jensen gap* $J_\pi := \mathbb{E}_\pi[U_o^{\text{drift}}] \cdot \mathbb{E}_\pi[1/U_o^{\text{drift}}] \geq 1$ is the strongest single empirical predictor of survival drop (Pearson $r = -0.57$, beating support-Kantorovich κ at -0.26, $\mathbb{E}_\pi[U_o^{\text{drift}}]$ at -0.23, and the IY-violation margin at -0.27). The worst-case Kantorovich bound $(1+\kappa)^2/(4\kappa)$ is never violated across the sweep — empirically grounding the Appendix G.6 strengthened scalar/matrix relationship.

C.6 Anisotropic-drift extension and the $\Lambda \propto \mathcal{I}_{\min}^{p}$ refinement

Under anisotropic drift $Q_\rho = \mathrm{diag}(\sigma_x^2, \sigma_y^2)$ with $\sigma_y > 0$, the LMI controller's $\Lambda \propto \mathcal{I}_{\min}$ calibration acquires a small wall-axis component that, multiplied by `wall_extreme`'s large wall-axis FIM, can flip the argmax to the wall-aligned action. Empirical pattern: at $\sigma_y = 0$, LMI survives 100%; at $\sigma_y = 1.0$ with $\sigma_x \in \{1.5, 2.0\}$, LMI matches the scalar controller's `wall_extreme` failure (0% survival).

Replacing $\Lambda \propto \mathcal{I}_{\min}$ with $\Lambda \propto \mathcal{I}_{\min}^{p}$ for $p > p_{\text{crit}}(\sigma_x, \sigma_y)$ (Appendix A.6) recovers directional concentration. Empirical survival rates across $p \in \{1, 1.5, 2, 3\}$ at four (σ_x, σ_y) cells:

Table 6: Anisotropic-drift survival rates across exponent p. Empirical breakdown matches closed-form p_{crit}.

(σ_x, σ_y)	p_{crit}	$p = 1$	$p = 1.5$	$p = 2$	$p = 3$
$(1.5, 0.5)$	0.79	1.00	1.00	1.00	1.00
$(1.5, 1.0)$	2.05	**0.00**	**0.00**	**0.00**	0.76
$(2.0, 0.5)$	0.60	1.00	1.00	1.00	1.00
$(2.0, 1.0)$	1.14	**0.00**	0.76	0.76	0.76

The empirical breakdown matches the closed-form p_{crit} exactly. The 76% survival cap at $\sigma_y = 1.0$ reflects `drift_probe`'s residual y-axis FIM (0.25); a finer action set with a stronger drift-only probe would push this closer to 100%.

C.7 k_L gain sweep — survival-vs-reward Pareto frontier

At $\sigma_w = 1.0$ (the row where the chosen calibration $k_L = 500$ matches greedy's 16% failure), sweeping $k_L \in [500, 5 \times 10^5]$ gives:

Table 7: k_L gain sweep at $\sigma_w = 1.0$ — survival rate / mean reward.

k_L	survival	mean reward
500	16%	61.4
1000	100%	0.0
2000	100%	0.0
5000	100%	0.0
50000	100%	0.0

The transition between $k_L = 500$ and $k_L = 1000$ is sharp (no smooth tradeoff at this resolution). Trip-threshold sweeps and fine σ_w sweeps (full results in supplementary materials) confirm: the calibration governs the Pareto frontier; trip threshold has no effect at $\sigma_w = 1.0$. The chosen $k_L = 500$ errs toward reward extraction.

C.8 Variants we did not run

- **Off-diagonal** $R_o(a)$ — actions whose observation covariance is rotated relative to the eigenframe of Q_ρ — tests whether range-containment holds without the simultaneous-diagonalizability assumption.
- **Continuous action space** with parametric $R_o(\theta)$ — admits closed-form derivation of optimal Λ direction.
- **Transient** $\mathcal{I}_{\min}(P_t)$ — meta-controller that updates its LMI floor as P_t evolves; would tighten the steady-state-DARE-form caveat in Appendix H.4.

D Scalar Reduction Proof

We give the explicit argument that (6) reduces to (15) in 1D, complementing Appendix G.6.

In 1D, $H \in \mathbb{R}$, $R_o(a) \in \mathbb{R}_{>0}$, and $\mathcal{I}_o(a) = H^2/R_o(a)$. WLOG set $H = 1$. The LMI (6) becomes the scalar information-yield inequality

$$\mathbb{E}_\pi[1/R_o(a)] \geq \mathcal{I}_{\min}, \tag{10}$$

with $\mathcal{I}_{\min}$ determined by the 1D DARE (Appendix A.3): $\mathcal{I}_{\min} = q/(R^2(R^2 - q))$ in the additive-drift limit with per-step drift variance q.

Identifying $U_o(a) = R_o(a)$ and $\mathrm{CIY}(a) = 1/U_o(a)$, (10) is $\mathbb{E}_\pi[\mathrm{CIY}(a)] \geq \mathcal{I}_{\min}$, which is the *information-yield* form of the 1D survival constraint. This form is *not* equivalent to the *observation-noise expectation* form $\mathbb{E}_\pi[U_o(a)] \leq U_o^{\max}$ used in Appendix F unless the action distribution is degenerate; by Jensen's inequality the two forms can diverge sharply for nondegenerate π. The scalar Appendix F derivation should therefore be read as the observation-noise-penalty version of the survival regime, with the LMI reduction (10) as the rigorous information-yield version. Both characterize the same survival regime under the surrogate CIY $\propto 1/U_o$, and at the boundary of binding constraints they coincide.

The matrix multiplier Λ collapses to a scalar $\lambda \geq 0$, complementary slackness becomes $\lambda(\mathbb{E}_\pi[\mathrm{CIY}] - \mathcal{I}_{\min}) = 0$, and the action selection rule reduces to $a_t \in \arg\max_a[Q_O(a) + \lambda \cdot \mathrm{CIY}(a)]$, which is (19) with λ drawn from the survival-margin family of Definition 4.1. The matrix and scalar formulations agree in 1D up to the Jensen-gap distinction between the noise-expectation and information-yield versions of the constraint. □

D.1 Kantorovich converse (Appendix G.6 (ii) detail)

For policies with $U_o(a) \in [u_{\min}, u_{\max}]$ on $\mathrm{supp}(\pi)$ and dispersion $\kappa := u_{\max}/u_{\min}$, the Kantorovich/Pólya-Szegő inequality [48,49] gives $\mathbb{E}_\pi[U_o(a)] \cdot \mathbb{E}_\pi[1/U_o(a)] \leq \frac{(1+\kappa)^2}{4\kappa}$, so the information-yield bound (16') implies a *relaxed* observation-noise bound $\mathbb{E}_\pi[U_o(a)] \leq \frac{(1+\kappa)^2}{4\kappa} \cdot U_o^{\max}$. The slack factor equals 1 at $\kappa = 1$ and grows monotonically in κ; tight at two-point distributions π with $p = 1/2$ at the support endpoints (Kantorovich worst case).

D.2 Multi-D generalization, simultaneously-diagonalizable (Appendix G.6 (iv) detail)

When the action FIMs $\{\mathcal{I}_o(a) : a \in \mathrm{supp}(\pi)\}$ are simultaneously diagonalizable — the regime of Appendix H's empirical setup — the Kantorovich operator inequality $\mathbb{E}_\pi[\mathcal{I}_o] \cdot \mathbb{E}_\pi[\mathcal{I}_o^{-1}] \preceq \frac{(1+\kappa)^2}{4\kappa} \cdot I$ holds with $\kappa = \lambda_{\max}/\lambda_{\min}$ over the joint spectrum, by reduction to per-eigendirection scalar Kantorovich. Beyond simultaneous-diagonalizability the analogous trace-form inequality $\mathrm{Tr}(\mathbb{E}_\pi[\mathcal{I}_o] \cdot \mathbb{E}_\pi[\mathcal{I}_o^{-1}]) \leq (1+\kappa)^2/(4\kappa) \cdot n$ extends to non-commuting families (cf. [50]; [51] ch. 4); the rigorous reach is not load-bearing for any Appendix H claim.

E 1D Gaussian-Kalman DARE Shadow-Price Calculation

For the 1D Gaussian-linear setup most directly aligned with Appendix F — drift $\delta_{t+1} = \beta\,\delta_t + w_t$, $w_t \sim \mathcal{N}(0, \rho^2)$, and observation $y_t = \delta_t + v_t$, $v_t \sim \mathcal{N}(0, U_o)$ — the steady-state predicted variance T and posterior variance S satisfy the DARE

$$T^2 + T\big[(1-\beta^2)U_o - \rho^2\big] - \rho^2\,U_o = 0, \qquad S = \frac{T\,U_o}{T + U_o}. \tag{11}$$

E.1 Steady-state variance sensitivity (closed form)

The boundary $\delta_t \in \mathcal{B}_R$ admits two natural criticality readings: *predicted* $T < R^2$ (the form Appendix G, Appendix H, and Appendix A.3 use, since Σ_δ is the steady-state DARE *predicted* covariance) and *filtered* $S < R^2$ (read against the posterior). The variance sensitivity at each boundary admits a closed form (with $\beta = 1$ for clarity):

$$\boxed{\left.\frac{\partial T}{\partial U_o}\right|_{T=R^2} = \frac{\rho^2}{2R^2-\rho^2}\ \text{(predicted, valid for } \rho^2 < R^2\text{)}, \quad \left.\frac{\partial S}{\partial U_o}\right|_{S=R^2} = \frac{\rho^2}{2R^2+\rho^2}\ \text{(filtered, any } \rho > 0\text{)}.} \tag{12}$$

The predicted-form qualifier $\rho^2 < R^2$ is the *capacity-binding* regime where the survival boundary $T = R^2$ is reachable (i.e., the persistence inequality is feasible at criticality); the algebraic regularity domain is wider ($\rho^2 < 2R^2$, where the denominator stays positive), but for $\rho^2 \in [R^2, 2R^2)$ the steady-state predicted variance never reaches R^2 in the first place — there is no boundary at which to evaluate the sensitivity. Both forms are bounded above by 1 in their valid regimes (predicted: $\rho^2 \to R^2$ from below; filtered: $\rho \to \infty$). Numerical verification of the filtered form across $\beta \in \{0.95, 1.00, 1.20, 1.50, 2.00, 10.0, 100\}$ with $R = 2$, $\rho = 1$ ($\beta = 1$ matches (12) exactly; $\beta \neq 1$ via implicit differentiation of (11)): $\left.\frac{\partial S}{\partial U_o}\right|_{U_o^c} \in \{0.055,\ 0.111,\ 0.336,\ 0.563,\ 0.751,\ 0.990,\ 0.99996\}$.

E.2 The variance sensitivity is *not* a Lagrange multiplier

The closed form (12) is *not* a Lagrange multiplier of any program in the paper. It is a structural derivative of the DARE map: how the steady-state posterior (or predicted) variance changes per unit observation noise, evaluated at the survival boundary. To interpret it as a "shadow price" would require identifying the optimization program whose dual variable equals this quantity; no such program is exhibited.

The actual KKT multiplier on the (15) constraint is the LP slope of $V^*(b) := \sup_\pi \mathbb{E}_\pi[Q_O(a)]$ s.t. $\mathbb{E}_\pi[U_o(a)] \le b$ — piecewise-linear concave nondecreasing in b with breakpoints at support-switching values, slopes bounded by $\max_{a\neq a'} |Q_O(a) - Q_O(a')|/|U_o(a) - U_o(a')|$. **Finite throughout the feasibility interior.** Reformulating (15) as a discounted infinite-horizon CMDP with absorbing failure preserves multiplier finiteness, with the value scaling at most as $1/(1-\gamma)$ in the standard limits. Only the undiscounted infinite-horizon limit produces a multiplier divergence, and only at the safe-policy feasibility boundary — a generic undiscounted-CMDP pathology, not a survival-specific phenomenon.

The numerical sweep on the long-run DP with absorbing failure at $|\delta| = R$ delivers a *positive* structural finding: the value gradient $\partial V^*/\partial R$ exhibits **gradient attenuation** at margin closure rather than divergence. With $\rho = 1$ and $R \in \{1.5, 2.0, 2.5, 3.0\}$, $V^*(0) \in \{7.4,\ 17.8,\ 37.4,\ 58.5\}$ and discrete slopes between successive R values $\partial V^*/\partial R \approx 21$ ($R : 1.5 \to 2.0$), 39 ($R : 2.0 \to 2.5$), 42 ($R : 2.5 \to 3.0$) — bounded above and growing only sub-linearly in R, so as $R \downarrow \rho^+$ (margin closing) the slope *decreases* rather than diverges. The interpretation is a Kramers-escape phenomenon: near the survival boundary the agent has effectively given up on probabilistic mass it cannot reach, so additional capacity buys *vanishing* marginal value — the opposite of barrier-function intuition. Combined with (12)'s closed-form local DARE sensitivity, this is independent evidence that the controller-side machinery has no divergence to chase.

E.3 Why this matters

Definition 4.1's divergent-gain construction is therefore best understood as a *controller-design overshoot* of the finite static multiplier — a deliberate choice ensuring robust constraint satisfaction under unmodeled disturbance, action-set discreteness, finite-sample noise, and the *finite, potentially small* worst-case exit-time window once the agent slips sub-threshold (Lemma 3.1 (ii)). The (12) closed form is intrinsically interesting as a measure of the dynamical sensitivity at the boundary — *one unit of observation-noise relaxation buys at most one unit of posterior-variance growth* — and it is the right quantity for understanding the local geometry of the steady-state DARE. But it is not the rate at which the agent values constraint relaxation; that rate is the LP multiplier, finite throughout.

F The Lyapunov-Survival Exploration Drive (Scalar)

This is the headline derivation. In the scalar case it is short; the multidimensional lift in Appendix G inherits its qualitative structure.

F.1 Coupling the Kalman gain to persistence

Consider an agent whose correction is the Kalman update with gain (1). In the scalar normalized setting — the regime Appendix F develops — the observation map has fixed sensitivity $c > 0$ *independent of the action-dependent observation noise* $U_o(a)$, and the bridge $\alpha = \eta^* c$ holds. *Derivation (scalar Kalman, $H = c$ a fixed channel constant).* The steady-state Kalman gain on prior state variance $U_M = P_-$ and observation noise U_o is $K = c\,U_M/(c^2 U_M + U_o)$, and the per-step closed-loop contraction on the mismatch $\delta = x - \hat{x}$ is $\alpha = Kc = c^2 U_M/(c^2 U_M + U_o)$. Rescaling $x \mapsto cx$, $U_o \mapsto U_o/c^2$ — the standard normalization — gives $\alpha = U_M/(U_M + U_o) = \eta^*$ with $c = 1$. The crucial property is that c is *independent of the action-dependent* U_o: the action sets observation noise but not channel sensitivity. In the multidimensional anisotropic setting where the observation map's directional sensitivity is not separable from $R_o(a)$, the scalar ceiling derivation breaks down, and the multidim treatment proceeds directly via the matrix Fisher information $\mathcal{I}_o(a) = H^\top R_o(a)^{-1} H$ in Appendix G rather than via a scalar bridge. Substituting into the structural persistence condition (5):

$$\eta^* c > \frac{\rho}{R} \quad \Longleftrightarrow \quad \frac{U_M}{U_M + U_o} > \frac{\rho}{Rc}. \tag{13}$$

Solving (13) for U_o gives the *survival-bound on observation noise*:

$$\boxed{U_o^{\max}(M_t) \;=\; U_M \left(\frac{R\,c}{\rho} - 1\right).} \tag{14}$$

The bound has three regimes in Rc versus ρ:

- **Feasible** ($Rc > \rho$). $U_o^{\max}(M_t) > 0$, proportional to U_M — the survival-noise ceiling collapses linearly as the agent's confidence grows. Probing the observation channel down to noise levels below the ceiling is necessary and sufficient for survival.
- **Threshold** ($Rc = \rho$). $U_o^{\max} = 0$ identically. Only the singular $U_o = 0$ peak-gain limit ($\eta^* = 1$) sits at the survival boundary, with no strict margin in U_o.
- **Infeasible** ($Rc < \rho$). $U_o^{\max} < 0$ — even zero observation noise cannot satisfy (13). The agent's *peak* correction rate is below the drift rate normalized by capacity. Survival in this regime is not rescuable by exploration alone; it requires improving the channel sensitivity c, the mismatch budget R, or the action class.

The infeasible regime is a *sharper* structural obstruction than the feasible-regime ceiling collapse — exploration cannot save an agent whose model class is below the drift-vs-capacity threshold. The remainder of Appendix F develops the feasible-regime story; the infeasible regime is named in main-text scope statements and not separately analyzed.

F.2 The KKT Lagrangian and survival-margin controller family

Restate the agent's policy problem. Let $Q_O(a)$ be the agent's pragmatic objective (expected reward or value). The agent maximizes Q_O subject to the survival constraint (14) on observation-noise expectation:

$$\max_{\pi} \; \mathbb{E}_{a\sim\pi}[Q_O(a)] \quad \text{s.t.} \quad \mathbb{E}_{a\sim\pi}[U_o(a)] \le U_o^{\max}(M_t). \tag{15}$$

The KKT Lagrangian for (15) introduces a non-negative shadow price $\lambda'(M_t) \ge 0$:

$$\mathcal{L}(\pi, \lambda') = \mathbb{E}_\pi[Q_O(a)] - \lambda' \left(\mathbb{E}_\pi[U_o(a)] - U_o^{\max}\right). \tag{16}$$

The KKT shadow price λ' is the marginal value of relaxing the constraint:

$$\lambda'(M_t) \;=\; \frac{\partial V^*(M_t)}{\partial U_o^{\max}(M_t)}, \tag{17}$$

where $V^*(b) := \max_\pi \mathbb{E}_\pi[Q_O(a)]$ s.t. $\mathbb{E}_\pi[U_o(a)] \le b$. As a finite-action LP over policies, V^* is concave, nondecreasing, piecewise-linear in b with breakpoints at support-switching values; its derivative $\lambda'(b)$ exists almost everywhere as the LP slope and is bounded above by $\max_{a \ne a'} |Q_O(a) - Q_O(a')|/|U_o(a) - U_o(a')|$ on the feasibility interior — finite and computable. In the 1D Gaussian-Kalman DARE setting (Appendix E), the *steady-state covariance sensitivity* $\partial S/\partial U_o$ at the survival boundary is independently bounded above by 1, with closed form

$$\left.\frac{\partial S}{\partial U_o}\right|_{\text{bdy}} = \frac{\rho^2}{2R^2 + \rho^2} \text{ (filtered) or } \frac{\rho^2}{2R^2 - \rho^2} \text{ (predicted)}, \tag{18}$$

saturating at 1 in the rapid-drift limit. (18) is *not* the LP multiplier on (15); it is a structural property of the DARE map — the geometric rate at which posterior covariance changes per unit observation noise — and composes with the value function only when an explicit long-horizon reward-stream dependence on S is written down. The naïve "barrier-function" intuition that $\lambda' \to \infty$ as the persistence margin closes is incorrect on the controller side: the static-program multiplier is bounded, and discounted infinite-horizon CMDP reformulations of (15) preserve multiplier finiteness (with $1/(1-\gamma)$ scaling). What does diverge is on the *environment side*; we return to this in Appendix F.5.

Definition 4.1 (stated in Section 4) names the family of policy weights that diverge at margin closure. Two natural family members illustrate the structure:

- **Inverse-uncertainty form**: $\lambda^{(1)}_{\text{surv}}(M_t) = k/U_M$ (with appropriate clamp for $U_M \to 0$). Diverges as $U_M \to 0$, which is when $U_o^{\max} \to 0$ and the boundary closes.
- **Margin-triggered form**: $\lambda^{(2)}_{\text{surv}}(M_t) = k\, U_M/(R - R^*)$ where $R^* = \rho/\alpha$ is the steady-state mismatch bound under the agent's currently-chosen action. Diverges as $R^* \uparrow R$, which is the persistence-margin closure event in the mismatch-dynamic coordinate.

Definition 4.1's divergence property is a *controller-design choice* — a deliberate overshoot of the bounded static-program multiplier (and bounded DARE variance sensitivity (18)) ensuring robust constraint satisfaction under unmodeled disturbance, action-set discreteness, and finite-sample noise. The reason to *want* this overshoot is environment-side, not multiplier-side: the controller's response loop must close inside the worst-case exit time T_{exit} of Lemma 3.1 (ii), which can be small once the agent's effective α drifts deep below threshold. Family membership ($\lambda_{\text{surv}} \to \infty$ at margin closure) is a *qualitative* divergence property; whether a chosen family member then *enforces* threshold (14) requires four explicit conditions — strict feasibility (F), a drift-aligned bonus (A; trivial in 1D, non-trivial in the multidimensional lift of Appendix G), gain dominance (G; implied by family membership in 1D), and argmax expressibility (P) — under which Proposition A.4 in Appendix A.7 establishes feasible-action selection. The inverse-uncertainty form anticipates the boundary via the agent's confidence; the margin-triggered form responds to the dynamic margin directly. Operationally, an additional safety factor $\gamma > 1$ replaces the per-step LMI threshold with $\gamma \cdot \mathcal{I}_{\min}$ (Appendix A.6); the Appendix H simulation witnesses pathwise survival at $\gamma \ge 50$.

F.3 From U_o to information yield

Bridge to standard information-yield formulations via $\text{CIY}(a) \propto 1/U_o(a)$ (causal information yield = action-conditional Fisher information of the observation channel) — *exact* in 1D Gaussian-linear since $\mathcal{N}(\mu, U_o)$'s Fisher information w.r.t. μ equals $1/U_o$, heuristic outside that regime (Section 6); the Appendix G matrix derivation identifies CIY directly with the FIM and removes the surrogate. A monotone information-yield surrogate to (15) is the value-plus-information objective:

$$\pi^* = \arg\max_\pi \mathbb{E}_\pi[Q_O(a) + \lambda_{\text{surv}}(M_t) \cdot \text{CIY}(a)], \tag{19}$$

with λ_{surv} drawn from Definition 4.1, and Proposition A.4 giving enforcement under F-A-G-P.

F.4 The dual-drive picture

The standard epistemic exploration drive in active inference, intrinsic motivation, and Bayesian experiment design carries a positive-with-uncertainty multiplier $\lambda_{\text{info}} \propto U_M$: explore *more* when uncertain. The survival multiplier (any member of the family in Definition 4.1) carries the *opposite* qualitative sign in U_M: it grows as the persistence margin closes, which (via $U_o^{\max} \propto U_M$) typically happens at *low* U_M in drifting environments. The two drives coexist:

$$\pi^* = \arg\max_{\pi} \mathbb{E}_\pi\big[Q_O(a) + \lambda_{\text{info}}(U_M)\,\text{CIY}(a) + \lambda_{\text{surv}}(M_t)\,\text{CIY}(a)\big], \tag{20}$$

with $\lambda_{\text{info}} \propto U_M$ dominating at high uncertainty (the standard regime; the agent learns) and λ_{surv} dominating near the persistence boundary (the regime where standard pictures predict pure exploitation; the agent must instead probe to track drift). The two drives are opposite-signed in U_M across the family.

F.5 The structural irony

The "tragedy" the paper's title points to is structural: in the standard exploration-exploitation contract, the agent explores when uncertain and exploits when confident; in a drifting environment, the persistence inequality (5) coupled to the Bayesian update gain (1) reverses this — *exploration is forced precisely at maximum confidence* ($U_M \to 0$), where the survival-noise budget $U_o^{\max}$ collapses to zero. A confident agent in a drifting world is, structurally, the most acutely-constrained: its small Kalman gain leaves it unable to track drift unless it actively probes for pristine observations.

Definition 4.1's divergent gain is a *controller-design choice* — a deliberate overshoot of the finite static-program multiplier on (15) that ensures robust constraint satisfaction under unmodeled disturbance, action-set discreteness, and finite-sample noise. It must live at the controller-design layer for a structural reason: Definition 4.1's divergence is in *state-space coordinates* — $\Delta(M_t) \to 0$, where the persistence margin is an agent-trajectory quantity computed from the realized filter — while any Lagrange multiplier on (15) lives in *parameter-space coordinates* over $b = U_o^{\max}$, which is a constraint-parameter quantity. These are limits over different spaces with no chain rule connecting them, so no constraint-parameterization choice for (15) could produce a parameter-space multiplier with Definition 4.1's state-space-shaped divergence. The reason to *want* the overshoot is environment-side, not controller-side: the worst-case adversarial exit time $T_{\text{exit}}(\delta_0)$ from Lemma 3.1 (ii) is *infinite above threshold* (no admissible disturbance forces exit) and *finite below threshold*, with floor of order R/ρ deep sub-threshold; the upper bound (4) becoming vacuous as $\bar{\alpha} \to (\rho/R)^-$ reflects safety-in-the-limit, not danger. The hazard is the *cliff* in the worst-case guarantee structure — $T_{\text{exit}} = \infty$ above threshold, finite the moment the agent slips below — leaving the controller's response loop a *finite, potentially small* window in which to close. The divergent-gain family enforces feasibility by construction even when the underlying multiplier is finite; the structural irony is that the agent's *controller* response is mathematically clean and finite-rate-derivable, but the *environment's clock* on the agent's mistakes is what closes the survival window. The long-run survival-truncated DP further reinforces this relocation: $\partial V^*/\partial R$ *attenuates* as the persistence margin closes (Appendix E.2), the opposite of barrier-function intuition — divergence is exclusively on the environment-side clock.

G The Blank-Wall Attack and the LMI Lift

The scalar derivation in Appendix F has a structural failure that the multidimensional setting exposes. Once exposed, the failure is also straightforward to fix using standard convex-analysis machinery, yielding a sharper version of the dual-drive structure.

G.1 The blank-wall attack

Consider an n-dimensional environment in which the drift acts only on a k-dimensional subspace $S_\rho \subset \mathbb{R}^n$. The constraint (15) bounds *scalar* expected observation noise $\mathbb{E}_\pi U_o(a) \le U_o^{\max}$ — it cares about magnitude, not alignment.

A pathological policy: choose actions whose observation channel is aligned *orthogonal* to S_ρ (e.g., observing a constant signal — a "blank wall" — that carries no information about the drifting coordinates). Such actions can drive scalar U_o arbitrarily close to zero, satisfying (15) trivially. Meanwhile, the agent's mismatch in S_ρ accumulates without bound because observations carry no information about it. The agent dies while satisfying the math.

This is the multidimensional analog of the bursting result of Anderson 1985 [9] under low persistent excitation: the agent has *some* excitation (observation channel is informative *in some directions*) but the wrong excitation (none in the drifting subspace).

G.2 Multidimensional setup

Following the linear-Gaussian information-form Kalman setup [45], let $\delta_t \in \mathbb{R}^n$ evolve as

$$\delta_{t+1} = (I - K_t H)\delta_t + w_t - K_t v_t(a_t), \tag{21}$$

with $w_t \sim \mathcal{N}(0, Q_\rho)$, $v_t \sim \mathcal{N}(0, R_o(a_t))$, optimal Kalman gain $K_t = P_t H^\top (H P_t H^\top + R_o(a_t))^{-1}$, steady-state mismatch covariance Σ_δ satisfying the DARE, and matrix-survival bound $\lambda_{\max}(\Sigma_\delta) < R^2$. The action-conditional Fisher Information Matrix $\mathcal{I}_o(a) = H^\top R_o(a)^{-1} H$ — *exactly* the matrix-valued causal information yield in the Gaussian-linear setting — enters the information-form Kalman update additively: $P^{-1}_{t+1|t+1} = P^{-1}_{t+1|t} + \mathcal{I}_o(a_t)$.

G.3 The LMI survival constraint

Theorem 4.2 (stated in Section 4) gives a sufficient condition on the averaged observation FIM for the deterministic DARE to admit a stabilizing steady-state solution with $\lambda_{\max}(\Sigma_\delta) < R^2$. Here we analyze its strength gradient and pathwise behavior.

Theorem 4.2 controls the deterministic DARE under expected observation noise; it does *not* by itself control the stochastic covariance trajectory P_t under realized action draws. The per-step Riccati update is nonlinear in $\mathcal{I}_o(a_t)$, and the expectation bound does not commute with the recursion. Lemma A.1 (Appendix A.5; matrix Chernoff via Tropp 2015 [47] Theorem 5.1.1 applied to iid PSD summands) lifts (6) to a pathwise high-probability lower bound on cumulative FIM at the rate $T\iota_{\min} - \mathcal{O}(\sqrt{T \log(k/\delta)})$ — but cumulative-information concentration does *not* translate into per-step covariance control: iid bursts of a near-degenerate-FIM action drawn with non-trivial probability mass have expected length $\Theta(\log T)$ and can drive $\lambda_{\max}(P_t) > R^2$ even when $\mathbb{E}_\pi[\mathcal{I}_o]$ exceeds $\mathcal{I}_{\min}$ by an arbitrary margin. Closing this gap requires per-action structure (Theorem A.2 / Theorem A.3 in Appendix A.6) or schedule assumptions stronger than iid. Sinopoli et al. 2004 [46] p. 4 flag the parallel averaged-vs-pathwise distinction in scalar-SISO intermittent Kalman.

(6) thus admits a natural strength gradient: Theorem 4.2's expectation form is the weakest member (deterministically sufficient on the DARE solution, pathwise sufficient on cumulative FIM); per-action LMI on the support — hypothesis F of Proposition A.4 applied to every $a \in \mathrm{supp}(\pi)$ — is the strongest, giving deterministic pathwise survival (Theorem A.2). The Appendix H LMI controller witnesses the strong end: per-action margin $\gamma \in [50, 960]$ across $\sigma_w \in [1.0, 3.0]$, so the *covariance recursion* is deterministically bounded by Theorem A.2; the observed 100/100 finite-horizon survival reflects this covariance bound combined with the action schedule's deterministic argmax. In the restricted setting (commuting $\{\mathcal{I}_o(a)\}$, scalar A, or i.i.d. stationary actions) the converse of (6) holds with per-eigendirection adversarial witness $w^*_k(t) = \rho_k\, \delta_k(t)/|\delta_k(t)|$ lifting Lemma 3.1 (iii); detail in Appendix A.4. The novel content of (6) is *interpretive*: $\mathcal{I}_{\min}$ is a *survival floor on the agent's policy*, not a designer's information budget — the agent must source enough information in the directions where drift acts to keep the steady-state covariance bounded.

G.4 The matrix Lagrangian and complementary slackness

The constrained problem $\max_\pi \mathbb{E}_\pi[Q_O(a)]$ s.t. $\mathbb{E}_\pi[\mathcal{I}_o(a)] \succeq \mathcal{I}_{\min}$ has a positive-semidefinite matrix Lagrange multiplier $\Lambda \succeq 0$ [52] §5.9 with KKT primal feasibility $\mathbb{E}_\pi[\mathcal{I}_o(a)] \succeq \mathcal{I}_{\min}$, dual feasibility $\Lambda \succeq 0$, and complementary slackness $\mathrm{Tr}(\Lambda(\mathbb{E}_\pi[\mathcal{I}_o(a)] - \mathcal{I}_{\min})) = 0$. The optimal action selection rule is

$$a_t \in \arg\max_a \left[\, Q_O(a) \;+\; \mathrm{Tr}(\Lambda \cdot \mathcal{I}_o(a)) \,\right]; \tag{22}$$

the trace inner product $\mathrm{Tr}(\Lambda \cdot \mathcal{I}_o(a))$ is the matrix analog of the scalar exploration bonus $\lambda_{\text{surv}} \cdot \mathrm{CIY}(a)$.

G.5 Resolution of the blank-wall attack

Complementary slackness gives $\mathrm{Tr}(\Lambda^*(\mathbb{E}_\pi[\mathcal{I}_o] - \mathcal{I}_{\min})) = 0$, killing multiplier mass on strictly-slack directions but not on its own forcing $\mathrm{range}(\Lambda^*) \subseteq \mathrm{range}(\mathcal{I}_{\min})$. Range containment is therefore a *controller-design hypothesis* on the family member: under simultaneous-diagonalizability (commuting eigenframes for $\{\mathcal{I}_o(a)\}_{a\in\mathcal{A}}$ and Q_ρ, or restriction to the diagonal LMI block at the steady-state covariance basis), the matrix family member $\Lambda \propto \mathcal{I}_{\min}$ used in Appendix H satisfies $\mathrm{range}(\Lambda) \subseteq \mathrm{range}(\mathcal{I}_{\min})$ by construction (positive scalar multiple), with zero weight in non-drifting eigendirections of Q_ρ where $\mathcal{I}_{\min}$ contributes zero.

For a blank-wall action — high $\mathcal{I}_o(a)$ supported in a non-drifting eigendirection — the trace product $\mathrm{Tr}(\Lambda^*\,\mathcal{I}_o(a))$ evaluates to zero. *The agent receives no exploration bonus for the blank-wall action, even though it satisfies any scalar magnitude bound.* The matrix Lagrangian discriminates by direction, not magnitude. This is the precise mathematical content of "the agent must probe in the directions that matter."

Proposition 4.3 (stated in Section 4) formalizes this: under the range-containment hypothesis $\mathrm{range}(\Lambda) \subseteq \mathrm{range}(\mathcal{I}_{\min})$, any blank-wall action a_w satisfies $\mathrm{Tr}(\Lambda\,\mathcal{I}_o(a_w)) = 0$. *Proof.* By the blank-wall hypothesis, $\mathrm{range}(\mathcal{I}_o(a_w)) \perp \mathrm{range}(\mathcal{I}_{\min}) \supseteq \mathrm{range}(\Lambda)$, so each column of $\mathcal{I}_o(a_w)$ lies in $\mathrm{range}(\Lambda)^\perp$. Since Λ is symmetric PSD, $\mathrm{range}(\Lambda)^\perp = \ker(\Lambda)$, hence $\Lambda\mathcal{I}_o(a_w) = 0$ as a matrix product and $\mathrm{Tr}(\Lambda\,\mathcal{I}_o(a_w)) = 0$. □

Proposition 4.3 says the blank-wall action receives zero *survival* bonus, not that it is precluded from selection: a high-Q_O blank-wall action can still be chosen, but it does so on its pragmatic merits, not because it satisfies a survival mandate it does not in fact satisfy. The agent's policy must source the survival information *somewhere*, and Proposition 4.3 ensures that "somewhere" is the drifting subspace.

Concretely, the matrix lift of either §3.3 family member — for instance the margin-triggered form $\Lambda \propto \mathcal{I}_{\min}/(R^2 - \lambda_{\max}(P_t))$ used in Appendix H — satisfies hypothesis A of Proposition A.4 by construction (range containment $\mathrm{range}(\Lambda) \subseteq \mathrm{range}(\mathcal{I}_{\min})$ holds because Λ is a positive scalar multiple of $\mathcal{I}_{\min}$). Theorem 4.2 + Proposition 4.3 then deliver the multidimensional enforcement story: under F (DARE feasibility), A (matrix lift's range containment), G (the controller's gain overshoots the reward-gap-induced argmax-flip threshold, automatic in the divergent-gain limit), and P (deterministic argmax expressibility), the matrix family-member's argmax lies in $\mathcal{A}_F$ at every state where the persistence margin is sufficiently small.

G.6 Scalar reduction (consistency check)

In the 1D case (6) degenerates to the scalar inequality $\mathbb{E}_\pi[\mathcal{I}_o(a)] = \mathbb{E}_\pi[1/U_o(a)] \geq \mathcal{I}_{\min}$ (16'), the *information-yield* form, while Appendix F produces the *observation-noise expectation* form $\mathbb{E}_\pi[U_o(a)] \leq U_o^{\max}$. The two are Jensen-linked: by convexity of $1/u$, the Appendix F bound is unconditionally a strict sufficient condition for (16') (any Appendix F-passing policy passes the LMI in 1D), with exact equivalence on degenerate policies, on constant-U_o supports ($\kappa = 1$ where $\kappa := u_{\max}/u_{\min}$), and asymptotically as $\kappa \to 1^+$ (the Appendix H deterministic controllers operate in this exact regime). The Kantorovich/Pólya-Szegő converse with multiplicative slack $(1+\kappa)^2/(4\kappa)$ and the simultaneously-diagonalizable multi-D extension are recorded in Appendix D. The matrix LMI (6) is the rigorous version going forward; Appendix F's form is the Jensen-tighter sufficient version; Definition 4.1 and the dual-drive structure (20) survive the tightening, and Λ collapses to a scalar shadow price with matrix exploration bonus reducing to $\lambda_{\text{surv}} \cdot \mathrm{CIY}(a)$.

H Empirical Illustration

We empirically demonstrate the directional-discrimination claim of the LMI lift (Theorem 4.2, Proposition 4.3) in a 2D drifting environment with separable drifting and non-drifting subspaces — the simplest setting in which the blank-wall attack of Appendix G.1 can be expressed. A scalar margin-triggered controller using $\mathrm{Tr}(\mathcal{I}_o)$ as its bonus is captured by the blank-wall trap; a matrix margin-triggered controller using $\mathrm{Tr}(\Lambda \cdot \mathcal{I}_o)$ with directional Λ is not. The simulation is small (single CPU, ~30 seconds) and the code is in supplementary materials (`sim/variant_causal_ib_2d.py`).

H.1 Setup

The environment has one drifting axis and one stationary axis under Model S: $\Omega^{(0)}_{t+1} = \Omega^{(0)}_t + w_t, \quad w_t \sim \mathcal{N}(0, \sigma_w^2); \qquad \Omega^{(1)}_{t+1} = \Omega^{(1)}_t$. The agent maintains a 2D belief (M_t, P_t) updated by the information-form Kalman recursion; the mismatch ceiling is $R = 4$ on either axis (the episode terminates when $\max_k |\delta^{(k)}_t| > R$). We sweep $\sigma_w \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$ over 100 episodes of 200 steps each per controller per parameter value. All reported survival rates carry a Wilson 95% CI of approximately $\pm 4\%$ at $n = 100$; the 100% / 0% binary cells in Table 1 are statistically sharp at this resolution.

The action set has five actions with diagonal observation covariances $R_o(a)$:

Table 8: Action set for the 2D simulation.

Action	$R_o(a)$ diag	$\mathcal{I}_o(a)$ diag	Tr($\mathcal{I}_o$)	Reward
`drift_probe`	$(0.25, 4.0)$	$(4.0, 0.25)$	4.25	no
`wall_probe`	$(4.0, 0.25)$	$(0.25, 4.0)$	4.25	no
`wall_extreme`	$(16.0, 0.04)$	$(0.0625, 25.0)$	25.06	no
`diag_probe`	$(1.0, 1.0)$	$(1.0, 1.0)$	2.00	no
`exploit`	$(9.0, 9.0)$	$(0.111, 0.111)$	0.22	yes ($Q_O = 1$)

The trap is that `wall_extreme` has *higher* total Fisher information (Tr($\mathcal{I}_o$) $= 25.06$) than `drift_probe` (Tr($\mathcal{I}_o$) $= 4.25$), so a controller that uses Tr($\mathcal{I}_o$) as its scalar exploration bonus prefers `wall_extreme` — even though `wall_extreme` puts essentially all of its information on the *non-drifting* axis. This is the Appendix G.1 blank-wall attack as a discrete-action concrete instance.

H.2 Controllers

All three controllers select $a^* = \arg\max_a[Q_O(a) + \text{bonus}(a)]$ and differ only in bonus(a):

- **Greedy.** bonus$(a) = 0$. Pure pragmatic exploitation.
- **Scalar margin-triggered.** bonus$(a) = \lambda \cdot \text{Tr}(\mathcal{I}_o(a))$ with $\lambda = k_s \, \text{Tr}(P_t)/(R^2 - \lambda_{\max}(P_t))$. This is the natural scalar 2D lift of the margin-triggered family member from Definition 4.1 — direction-blind by construction.
- **LMI margin-triggered.** bonus$(a) = \text{Tr}(\Lambda \cdot \mathcal{I}_o(a))$ with $\Lambda = k_L \, \mathcal{I}_{\min}/(R^2 - \lambda_{\max}(P_t))$, where $\mathcal{I}_{\min}$ is the per-axis DARE-derived floor (Appendix A.3). In our setup $\mathcal{I}_{\min} = \text{diag}(\sigma_w^2/(R^2(R^2 - \sigma_w^2)), 0)$, so by construction range$(\Lambda) \subseteq$ range$(\mathcal{I}_{\min})$ — the precise range-containment hypothesis of Proposition 4.3.

We use $k_s = 1$, $k_L = 500$, calibrated so the exploration bonus is on the same order of magnitude as the exploit pragmatic value $Q_O = 1$.

H.3 Results

Table 9 summarizes the headline pattern across σ_w:

Table 9: Survival rate / mean reward across σ_w for greedy, scalar margin-triggered, and LMI margin-triggered controllers (100 episodes × 200 steps each).

σ_w	greedy survival / reward	scalar survival / reward	LMI survival / reward
0.5	95% / 147.5	71% / 0	95% / 147.5
1.0	16% / 61.4	1% / 0	16% / 61.4
1.5	0% / 25.6	0% / 0	**100% / 0**
2.0	0% / 16.5	0% / 0	**100% / 0**
2.5	0% / 11.7	0% / 0	**100% / 0**
3.0	0% / 8.5	0% / 0	**100% / 0**

The pattern admits three clean readings.

(i) The blank-wall attack is real and reproducible Across all σ_w, the scalar controller selects `wall_extreme` 100% of the time. At $\sigma_w = 2.0$, its mean accumulated FIM along the drift axis is 0.96 while along the stationary axis it is 383.25 — *two orders of magnitude more information gathered, all in the wrong direction*. The scalar controller dies even faster than greedy (mean survival 14.3 steps vs 25.0): `wall_extreme`'s observation channel on the drift axis is *noisier* than exploit's, so the agent is worse at tracking drift while supposedly probing.

(ii) LMI directional discrimination works as predicted When the LMI controller fires ($\sigma_w \geq 1.5$), it selects `drift_probe` 100% and never picks a wall-aligned action. Per-axis FIM at $\sigma_w = 2.0$: drift axis 800, stationary axis 50 (residual from `drift_probe`'s $\mathcal{I}_o[1,1] = 0.25$; the agent never *probes* the wall). With $\text{range}(\Lambda) = \text{range}(\mathcal{I}_{\text{min}}) = \text{span}(e_0)$, $\text{Tr}(\Lambda \mathcal{I}_o(a_w)) \approx \Lambda_{00} \cdot 0.0625 \approx 0$ for `wall_extreme` regardless of its huge $\mathcal{I}_o[1,1]$ — Proposition 4.3 instantiated empirically.

(iii) At low drift, LMI matches greedy At $\sigma_w \in \{0.5, 1.0\}$ the LMI controller's action histogram, survival rate, and reward are bit-identical to greedy: high survival at $\sigma_w = 0.5$, only 16% at $\sigma_w = 1.0$ for both. Under the chosen $k_L = 500$ calibration (a *reward-favoring* operating point) the survival drive's trigger remains silent at $\sigma_w = 1.0$ because $\mathcal{I}_{\text{min},0} \propto \sigma_w^2$ falls below exploit's pragmatic value $Q_O = 1$; raising $k_L \geq 1000$ recovers 100% survival at zero mean reward (Appendix C.7), tracing the survival/reward Pareto frontier. The bistable-gain artifact (Appendix H.4, Appendix B.5) prevents a graded tradeoff between the two ends.

Action-selection histograms (`sim/plots/fig2`) are deterministic; the per-axis FIM scatter (`fig4`) shows the three controllers as three disjoint clusters.

H.4 What this experiment does not show

We are explicit about the experiment's scope.

A smooth survival-vs-reward Pareto frontier is not achieved The matrix margin-triggered controller exhibits a sharp bistable transition in its gain k_L: at $\sigma_w = 1.5$, sweeping $k_L \in [10, 2000]$ shows survival rate jumps from 0% to 100% between $k_L = 300$ and $k_L = 400$, with mean reward correspondingly dropping from 25.6 to 0 (supplementary figure `fig5_lmi_bistability`). There is no intermediate k_L that produces graded tradeoff. Mechanistically this is the same artifact the 1D simulation's analogous binary-action setup produces (see Appendix B): in our action set, a single `drift_probe` step rapidly drives $P[0,0] \to 0.2$, which keeps the persistence margin near its maximum value $R^2 - \lambda_{\text{max}}(P_t) \approx 16$ permanently — the margin trigger never modulates dynamically. Smooth Pareto-optimality of the matrix lift remains an empirical open question; it likely requires stochastic policies, action sets where a single probe does not dominate steady-state tracking, or a different functional form for the trigger.

The action set is simultaneously diagonalizable Both $R_o(a)$ and Q_ρ are diagonal in the same basis, which is the regime where the LMI cleanly decouples into per-eigendirection scalar inequalities (Appendix A.4). Off-diagonal R_o or non-commuting action FIMs would test the converse-and-restriction story we record there but did not implement here.

The `wall_extreme` action is illustrative, not realistic. Its observation covariance $R_o = (16, 0.04)$ is engineered so the scalar-trace bonus strictly prefers it over `drift_probe` — making the failure mode legible in a 5-action discrete setting. In practice, the blank-wall structure arises wherever the observation channel can be steered orthogonally to drift: orientation-only sensors on a mobile platform whose position drifts; degenerate observation modalities in partially-observed control (e.g., sensors saturating in one direction while remaining sensitive in another); or any action class where some action provides high information about a stationary attribute and low information about the drifting state. The point of `wall_extreme` is to make the trap visible, not to be representative; less engineered action sets exhibit the same structure with the magnitudes attenuated.

The $\mathcal{I}_{\text{min}}$ used is the steady-state DARE form. Transient-regime $\mathcal{I}_{\text{min}}(P_t)$ would depend on the agent's current covariance; the steady-state form is a conservative bound that suffices for the directional-discrimination claim but does not characterize finite-time behavior optimally.

Stochastic-policy + calibration extensions (Appendix C.5–Appendix C.7). Softmax-randomized variants at low temperature ($T = 0.1$) match the deterministic controller's 100% survival; higher temperatures spread mass across non-drift-aligned actions and survival drops, with the policy's Jensen gap $J_\pi := \mathbb{E}_\pi[U_o^{\text{drift}}] \cdot \mathbb{E}_\pi[1/U_o^{\text{drift}}]$ as the strongest single empirical predictor ($r = -0.57$ across 20 variants; Kantorovich bound never violated). The chosen calibration $k_L = 500$ matches greedy at low drift; $k_L \geq 1000$ raises survival to 100% at zero mean reward — a Pareto frontier between survival and reward, with the operating point a free design parameter consistent with Definition 4.1's overshoot reading.

H.5 1D consistency check

A 1D toy version with two actions (exploit, explore) and the margin-triggered $\lambda = k\, U_M/(R - R^*)$ controller — kept as an illustrative consistency check rather than a separately reproduced experiment, with hyperparameters at Appendix B.3 — confirms the threshold's *bite*: strict-exploit greedy fails persistence; survival-margin family members preserve it. In 1D the blank-wall attack does not exist (no orthogonal-to-drift subspace), so the 2D experiment above is the rigorous test of the Appendix G LMI claims.

I Discussion

I.1 Two parallel exploration drives

The standard exploration-exploitation picture has one drive: epistemic, scaling with uncertainty. The dual-drive picture (20) has two drives, scaling oppositely in U_M:

- $\lambda_{\text{info}} \propto U_M$ — *epistemic* drive. Explore to learn. Dominates at high uncertainty.
- λ_{surv} from the survival-margin family (Definition 4.1) — *Lyapunov-survival* drive. Explore to stay alive. Diverges as the persistence margin $R - R^*$ closes, which (via $U_o^{\max} \propto U_M$) typically happens at low U_M in drifting environments.

Both reduce to zero in stationary environments ($\rho = 0$): $\mathcal{I}_{\min} \to 0$ when $Q_\rho \to 0$. The survival drive is therefore a *drift-specific* phenomenon. In the matrix lift (Appendix G) the two drives admit a structural decomposition $\Lambda = \Lambda_{\text{info}} + \Lambda_{\text{surv}}$ in which Λ_{surv} has support on eigendirections where the LMI binds — under simultaneous-diagonalizability, the drifting eigendirections where the agent has grown confident.

I.2 Comparison to persistent excitation

Isn't this just persistent excitation under a different name? No, in two respects. First, PE is a *positive condition on the input* — a richness criterion $\sum_{t=k}^{k+N} u_t u_t^\top \succeq \mu I$ [9,53] — a constraint on the *designer*, while the Lyapunov-survival result is a *derived bound on the policy's expected FIM*. PE is the vehicle for satisfying (6); not what (6) says. Second, (6) supplies a *directional* shadow price coupling $\mathcal{I}_{\min}$ to the drift covariance Q_ρ via the DARE, where typical PE conditions are isotropic (μI) or — in directional-forgetting work [16,17] — directional in the input but not coupled to a Lyapunov stability theorem.

I.3 Comparison to intrinsic motivation

Intrinsic-motivation methods (ICM [2], RND [4], VIME [3]) carry a positive-with-uncertainty bonus and share the standard monotonicity (confidence ⇒ exploitation); the Lyapunov-survival drive is a *complement* — silent in stationary settings, dominant near the persistence boundary in drifting ones.

I.4 Comparison to active-inference dark-room debate

The dark-room problem [6] — surprise-minimizing agents trivially satisfied by sensory deprivation — has a *parametric* reply [7] (priors encoding engagement preference) and a fragility result [8] (the epistemic component of EFE collapses to a state-independent constant in linear-Gaussian state-space models, leaving only the pragmatic prior-driven term — the parametric escape doing all the work). The Lyapunov-survival result provides a *structural* escape: a dark-room agent has $\mathbb{E}[\mathcal{I}_o(a)] = 0$ in the drifting subspace, violates (6), and loses bounded tracking — without needing any prior over outcomes. The structural escape is robust to the linear-Gaussian EFE-collapse precisely because it routes through the persistence inequality (5) rather than through the EFE epistemic term, and (5) is sharper (not weaker) in linear-Gaussian settings. Not a criticism of active inference: a relocation of the "explore-the-rich-environment" mandate from preferences-as-priors to Lyapunov mathematics.

I.5 Connection to dual control and probing

Dual control [11–13,21,22] motivates probing via a Bayesian regulation-vs-identification trade-off; the Lyapunov-survival result provides an *additional* persistence-based motivation — probing is required to maintain bounded tracking *now* under drift, not just to reduce parameter uncertainty for future control — and the resulting term is non-zero even at high confidence where the standard dual-control probing term is at its minimum.

I.6 Connection to Fisher-information experiment design

LMI-based closed-loop experiment design [14,23,54] and Tanaka et al. 2017 [15] (Gaussian sequential rate-distortion as an SDP with FIM constraints — closest existing work to our (6)) share the matrix-Lagrangian / DARE apparatus but use it differently: $\mathcal{I}_{\min}$ enters as a *survival floor on the agent's policy*, not as a designer's information budget, and Λ is a property of the agent's policy at equilibrium rather than a designer's choice variable.

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